Relational Separation Logic for Compiler Verification

Gabriel Desfrene
Supervised by Xavier Leroy and François Pottier

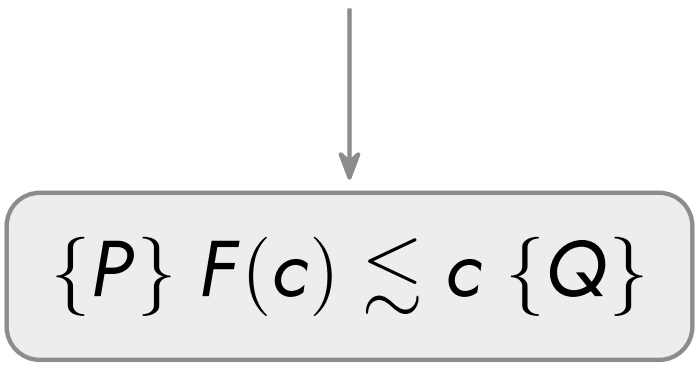
September 2026

Research Goals

$$\{P\}\ F(c) \lesssim c\ \{Q\}$$

Research Goals

Express compiler correctness?



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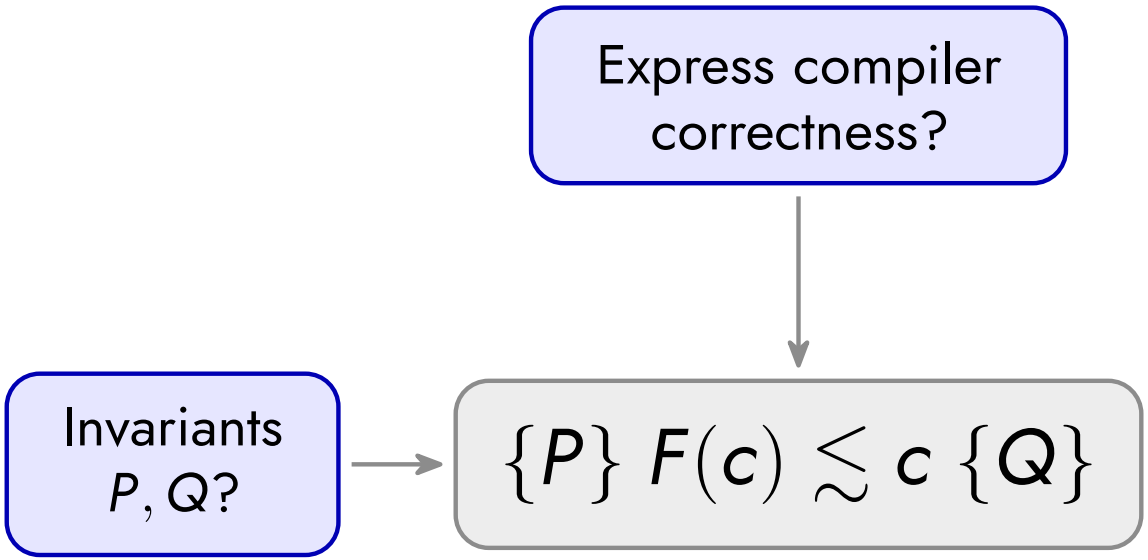

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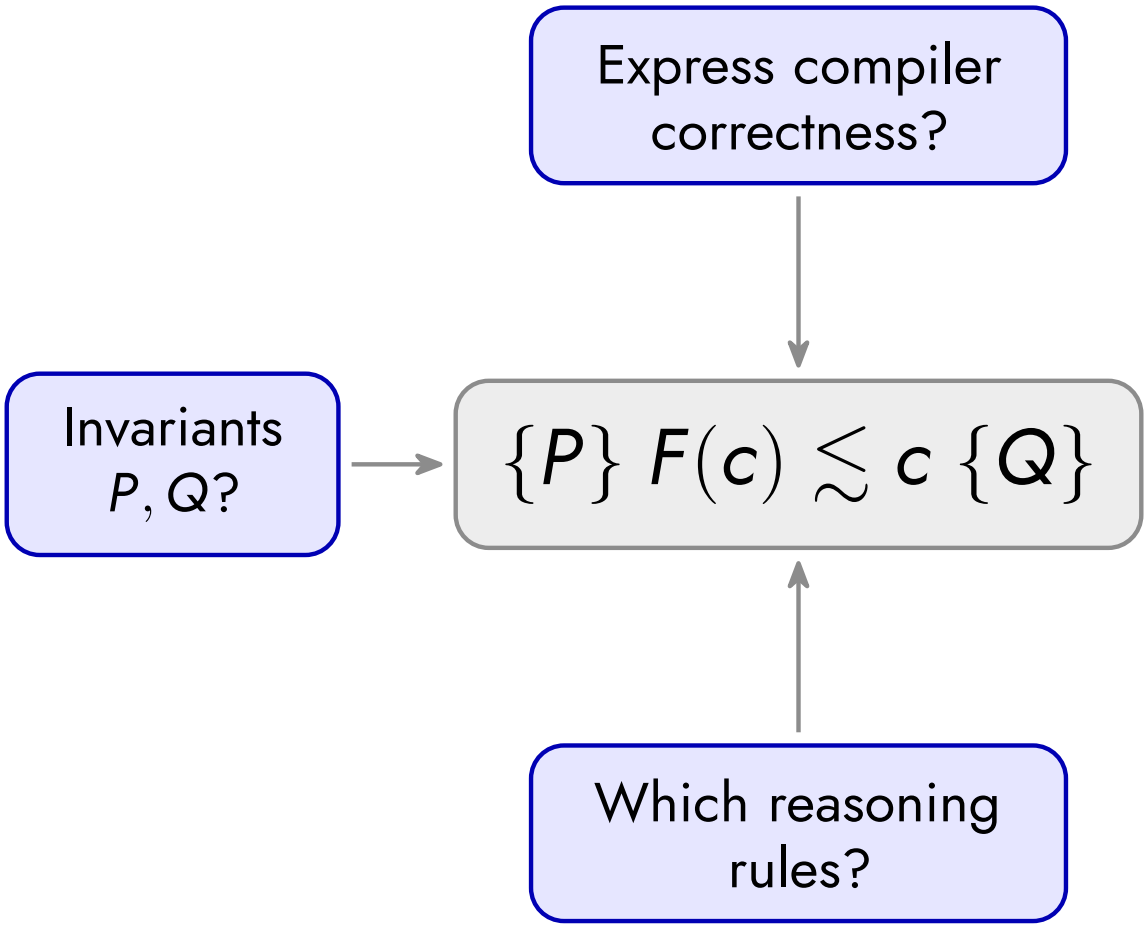

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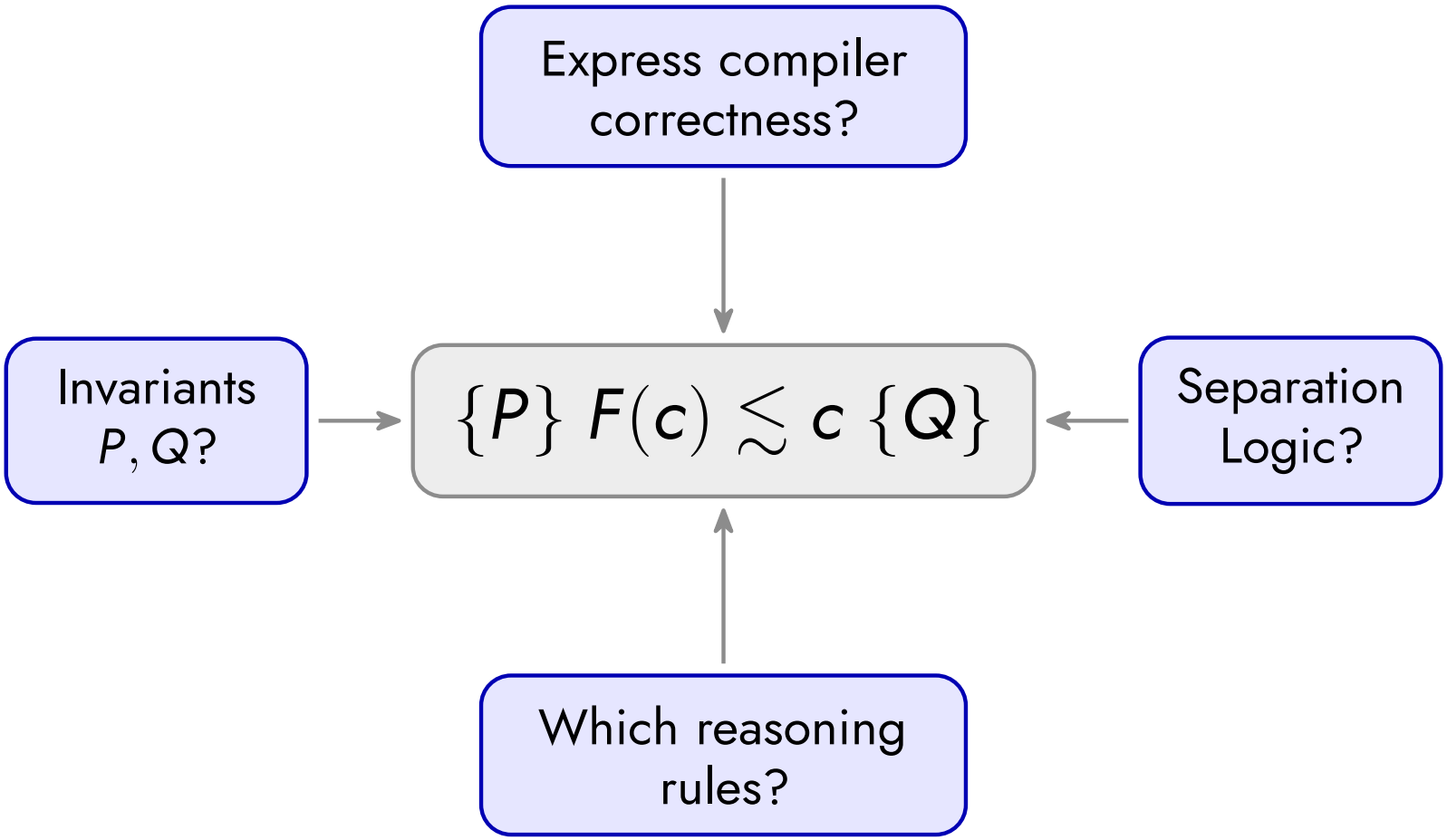

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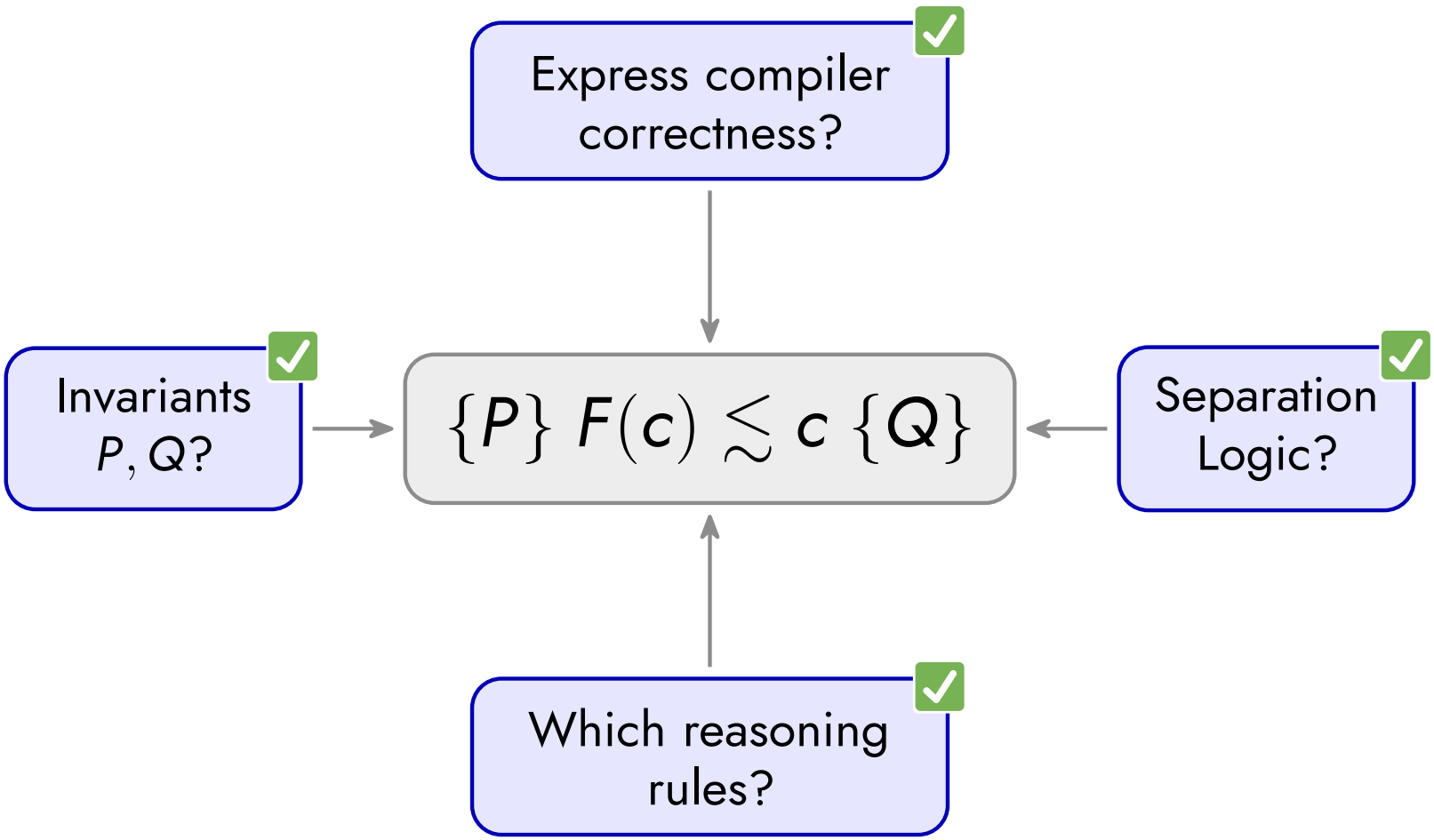

Context

Compilers Can Introduce Bugs

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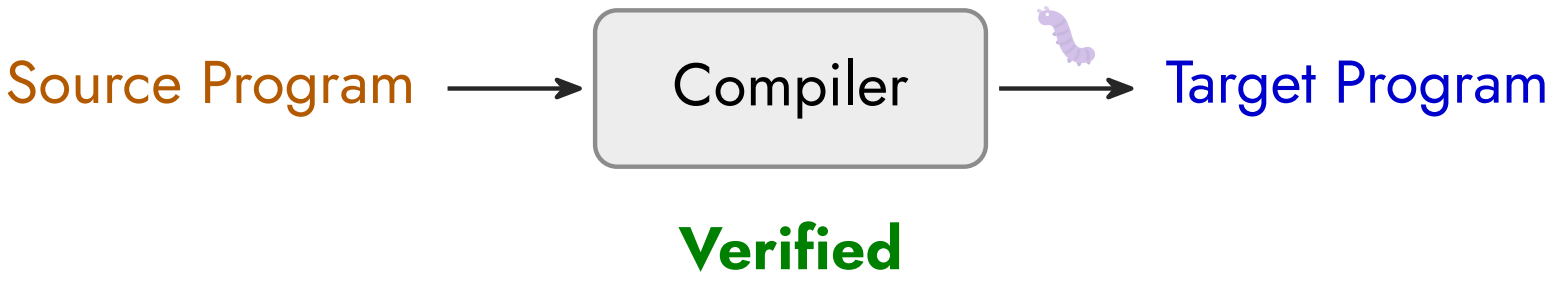

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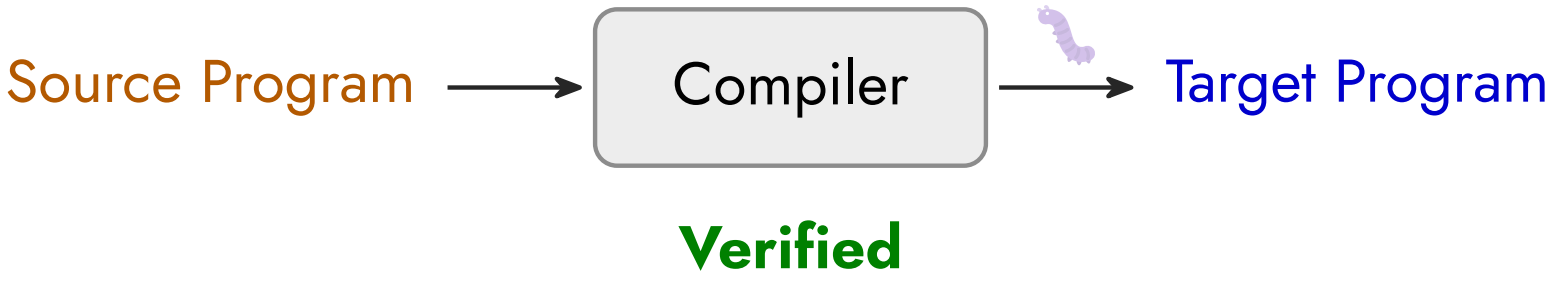

CakeML
ML → machine

CompCert
C → assembly

Jasmin
crypto → machine

Behavioral Refinement

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$$b \in \{\texttt{Term}(v, m) \mid \texttt{Div} \mid \lightning\}$$

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$b \in T \iff b \in S$

$$\mathtt{f}\big((\mathtt{print}(\texttt{"A"}); 1), (\mathtt{print}(\texttt{"B"}); 2)\big)$$
$$\downarrow$$
$$\mathtt{print}(\texttt{"A"}); \mathtt{print}(\texttt{"B"}); \mathtt{f}(1, 2)$$

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$T \preceq S \triangleq$

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$T \preceq S \triangleq$

$\lightning \in S$

or

$b \in T \implies b \in S$

Simulation Relation

$$t_0 \to t_1 \to \cdots \to \texttt{Term}(v, m)$$

$$\Downarrow$$

$$s_0 \to s_1 \to \cdots \to \texttt{Term}(v, m)$$

Simulation Relation

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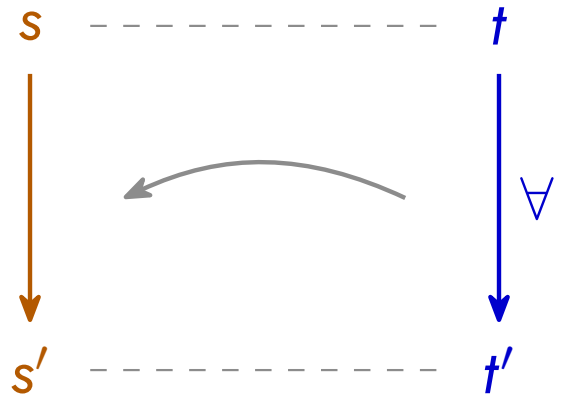

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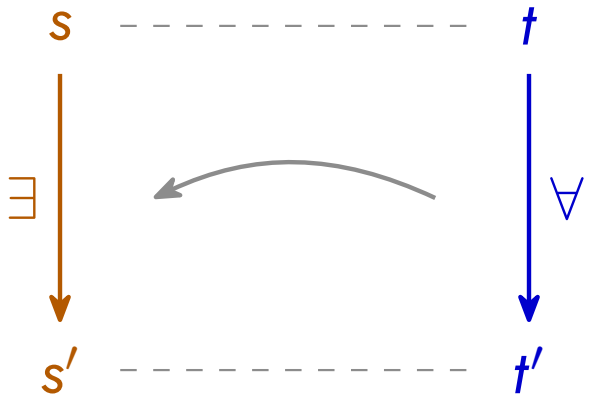
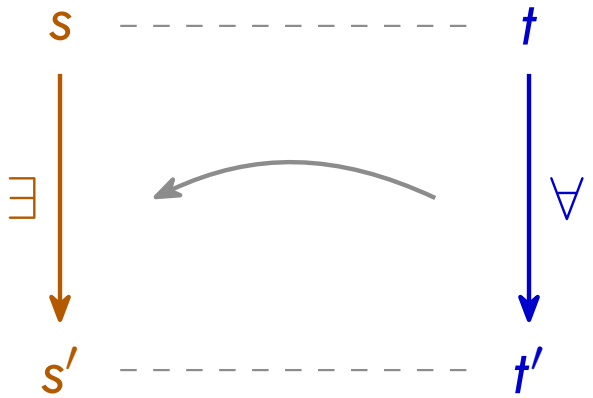


Simulation Relation

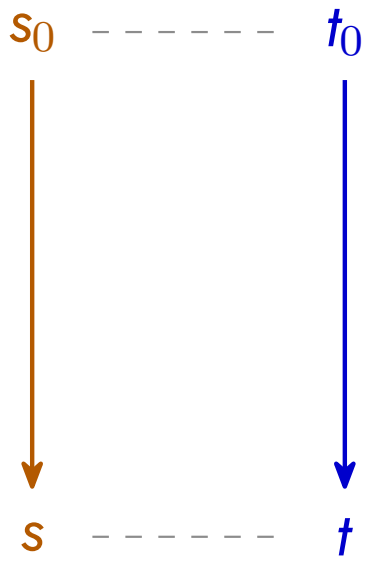
Backward Simulation



$$\Downarrow$$

$$T \preceq S$$

The Stuttering Problem



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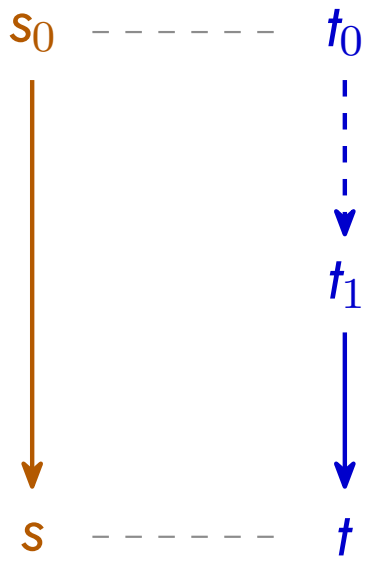

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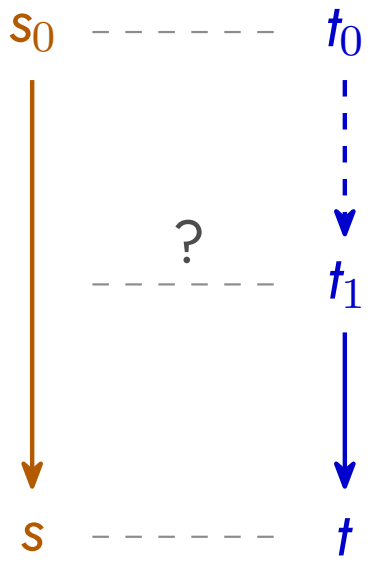

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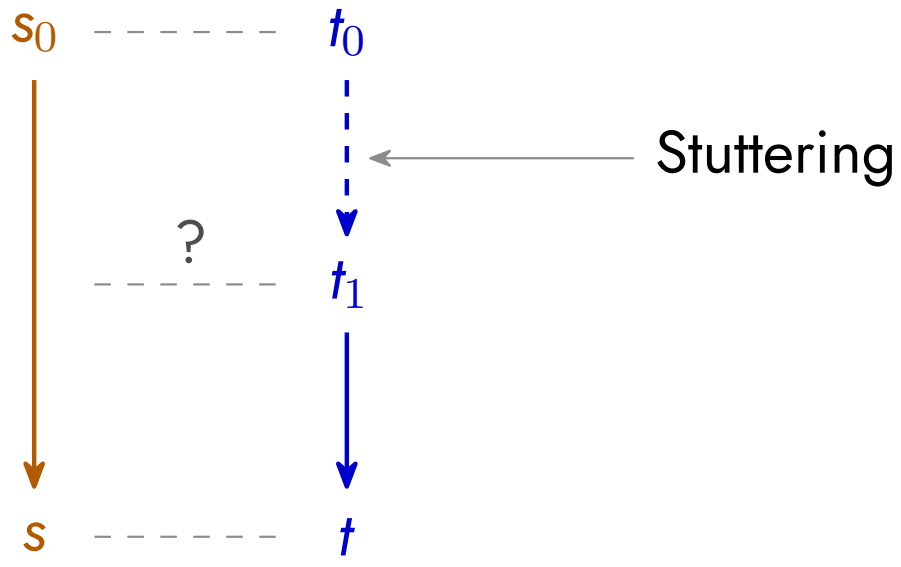

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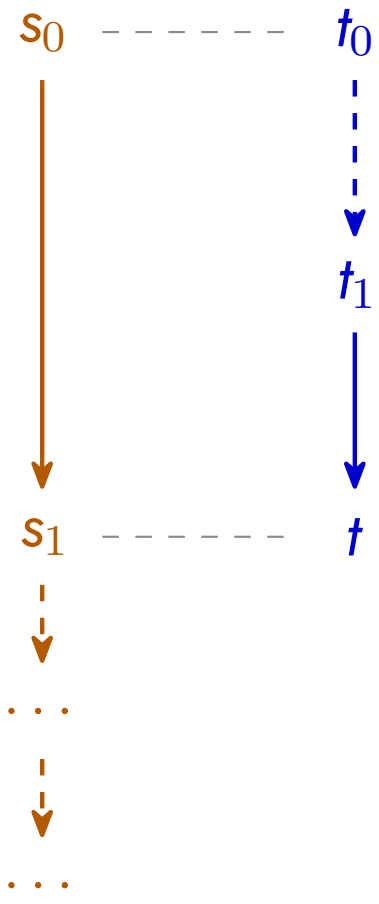

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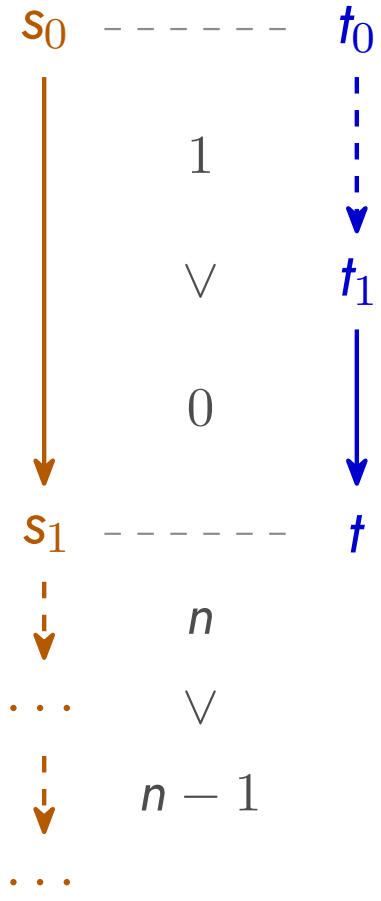

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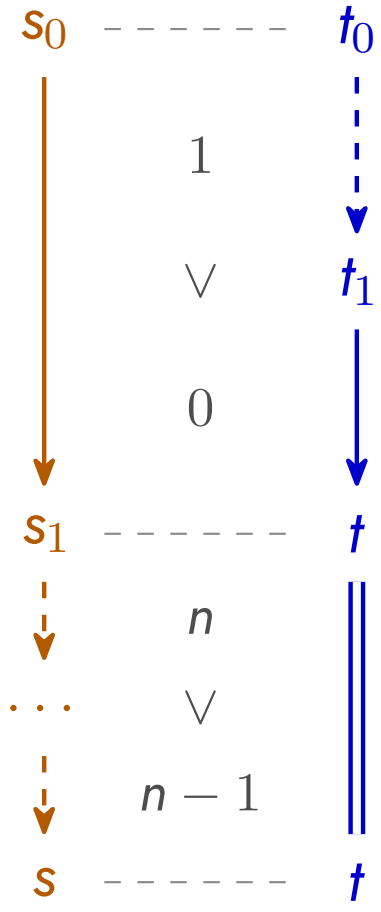

Limitations of Existing Approaches

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	Non-Determinism	

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	Backward Simulation	
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	Non-Determinism	Memory Reasoning
CakeML	Not Supported	
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Limitations of Existing Approaches

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CakeML	Not Supported	Global Invariants
CompCert	External Only	Global Invariants

Backward Simulation

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CakeML	Not Supported	Global Invariants
	Backward Simulation	**Separation Logic**
CompCert	External Only	Global Invariants

Contributions

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Backward Simulation

inspired by "Stuttering For Free"

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Relational Separation Logic

inspired by "Simuliris"

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Contributions

Language Dependent

Language Independent

Relational Separation Logic

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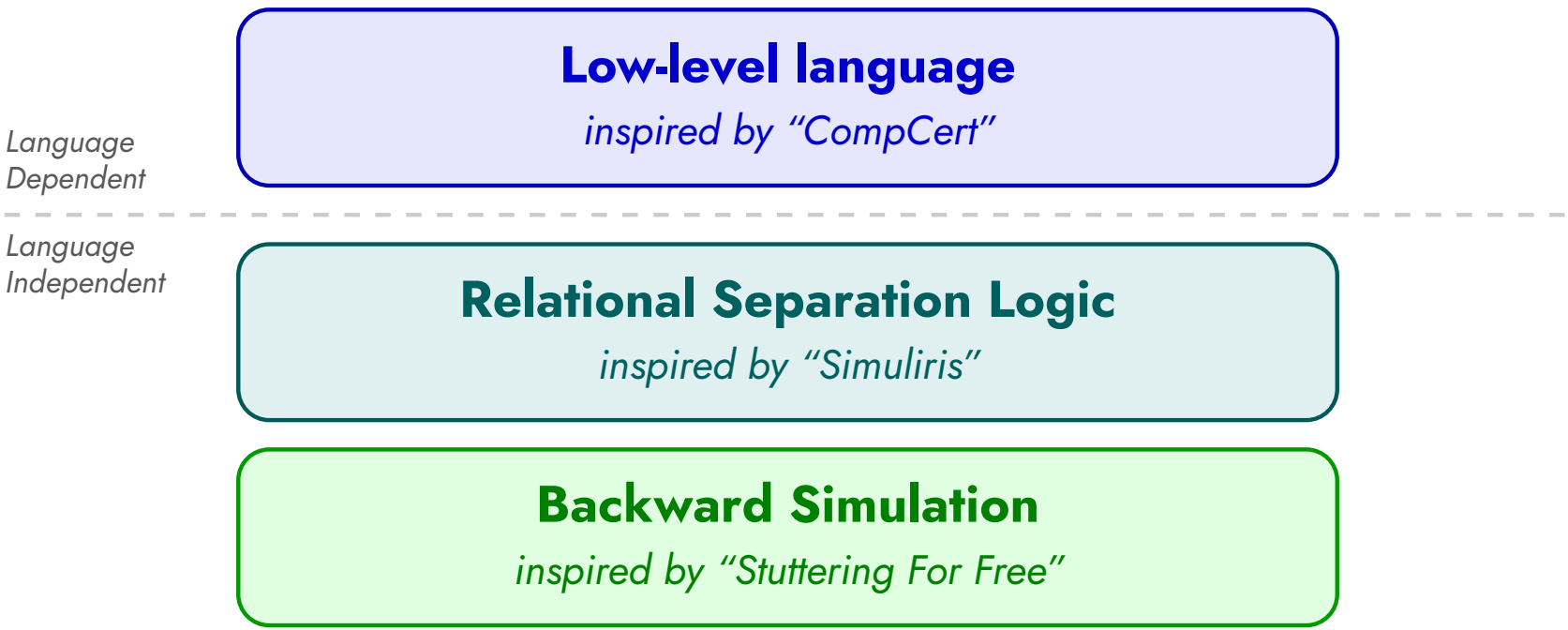

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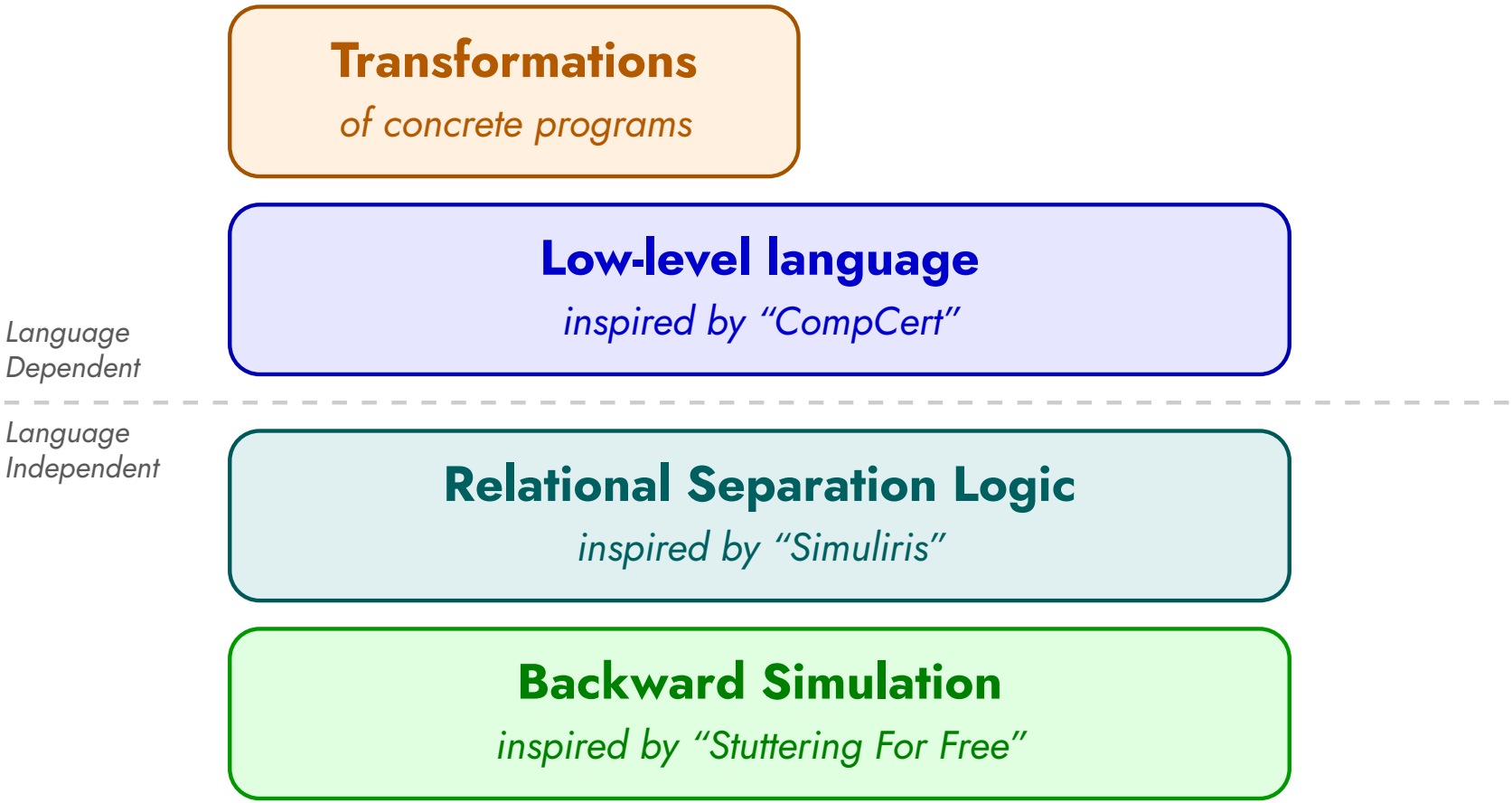

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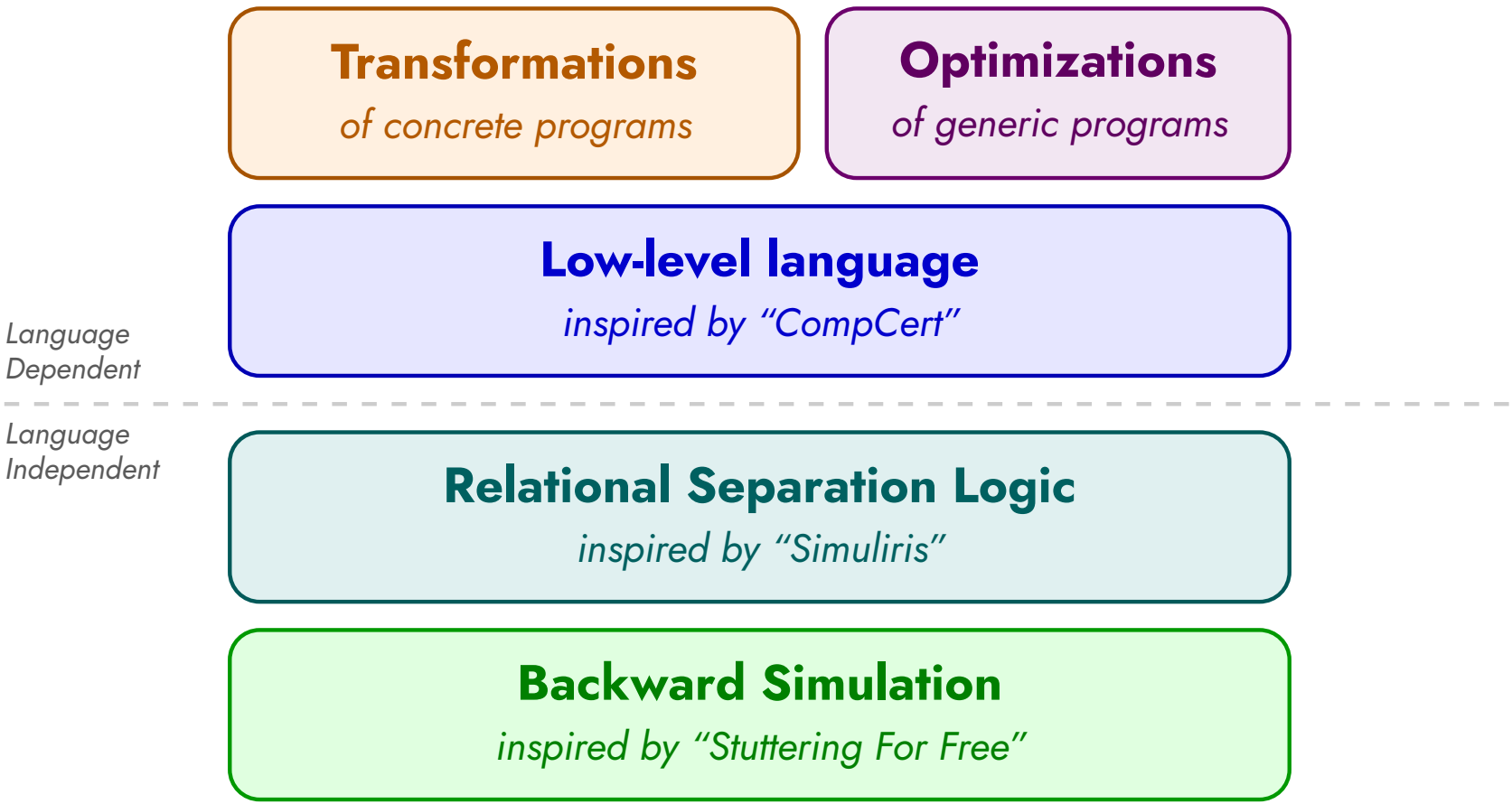

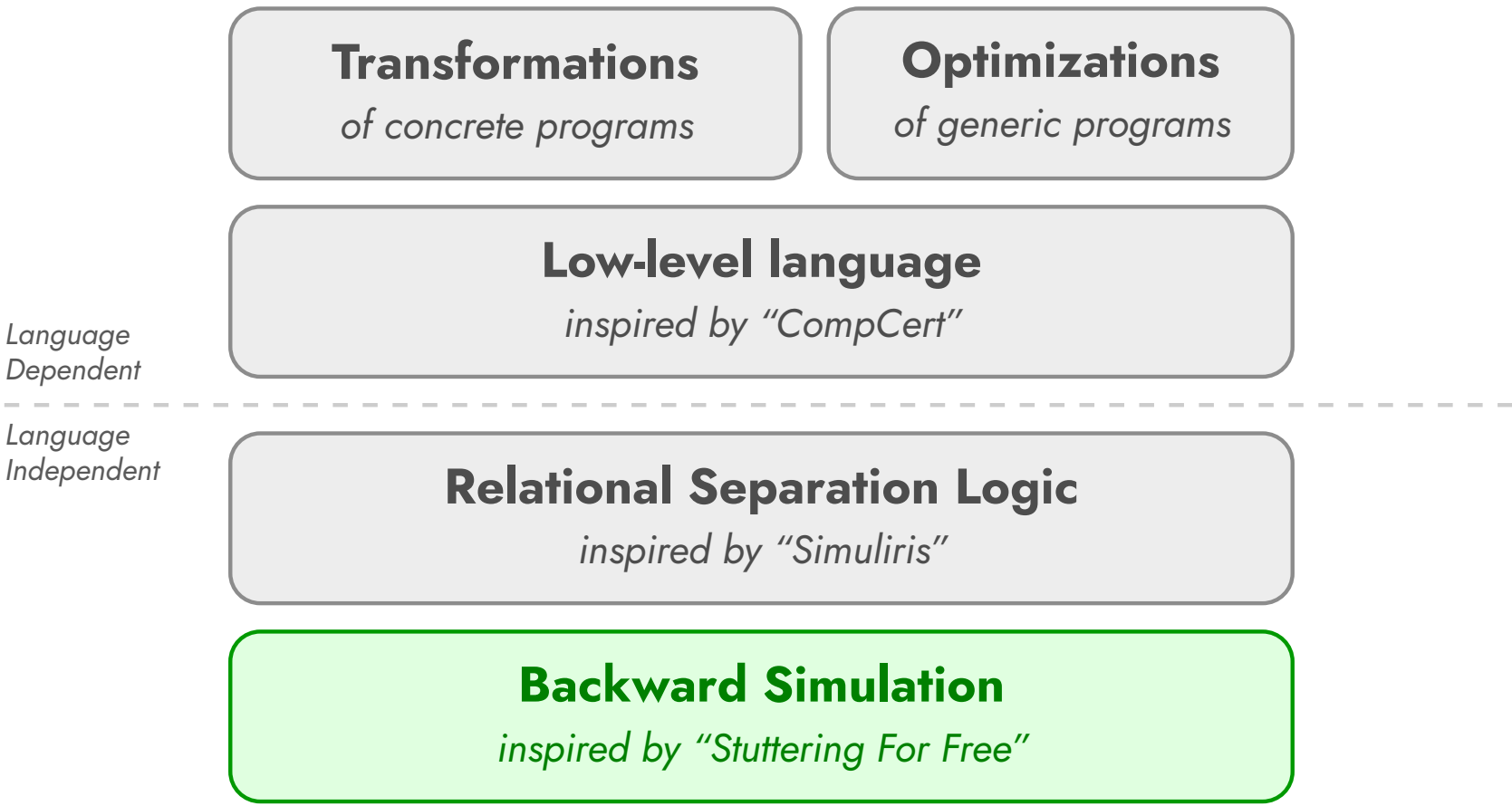
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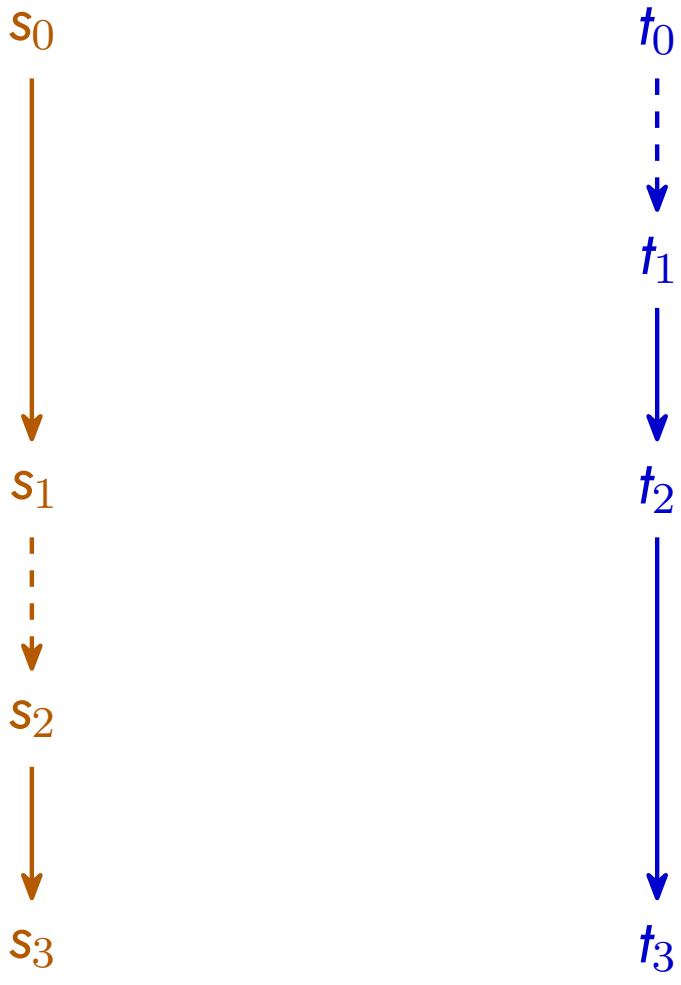

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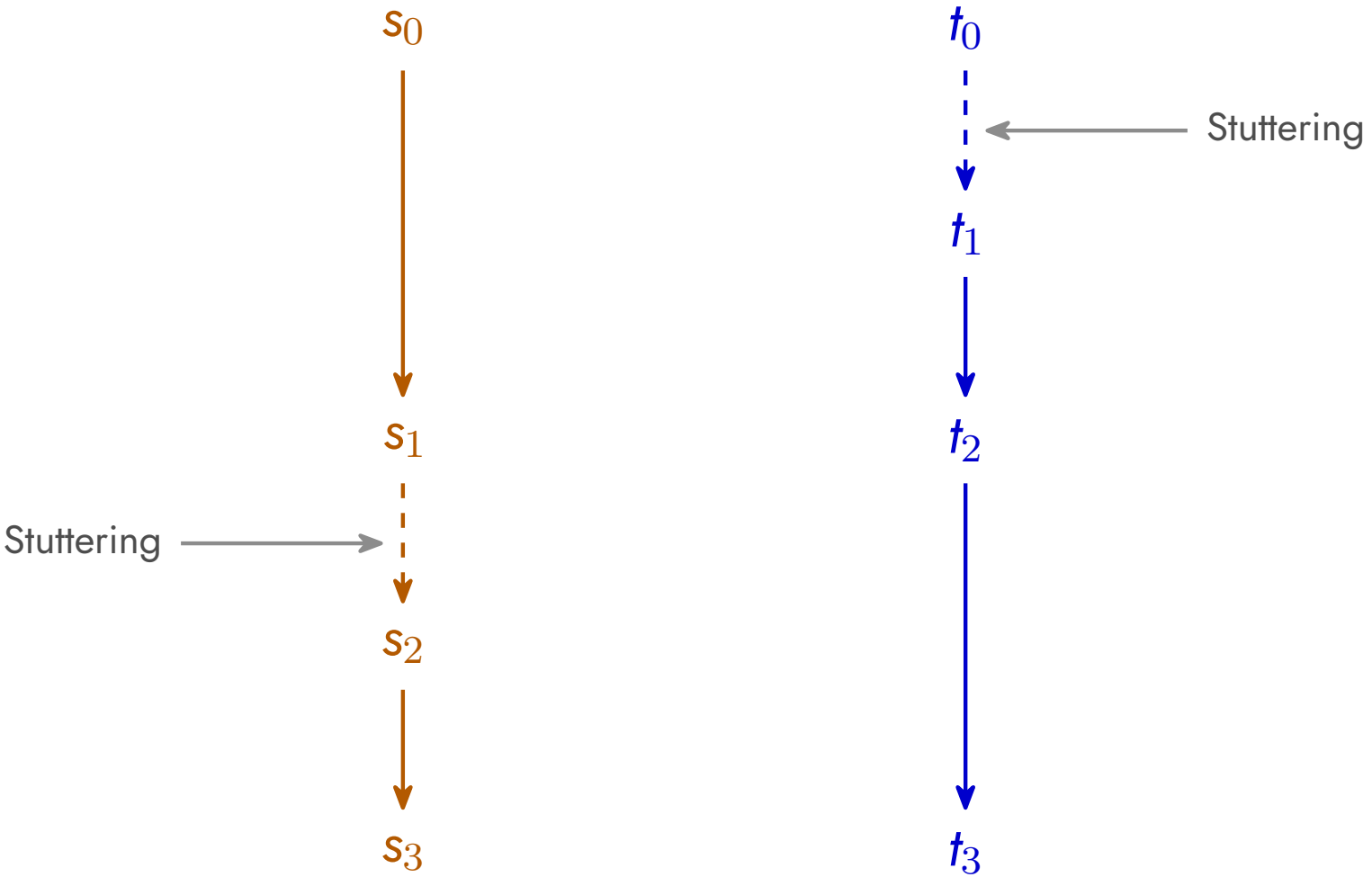

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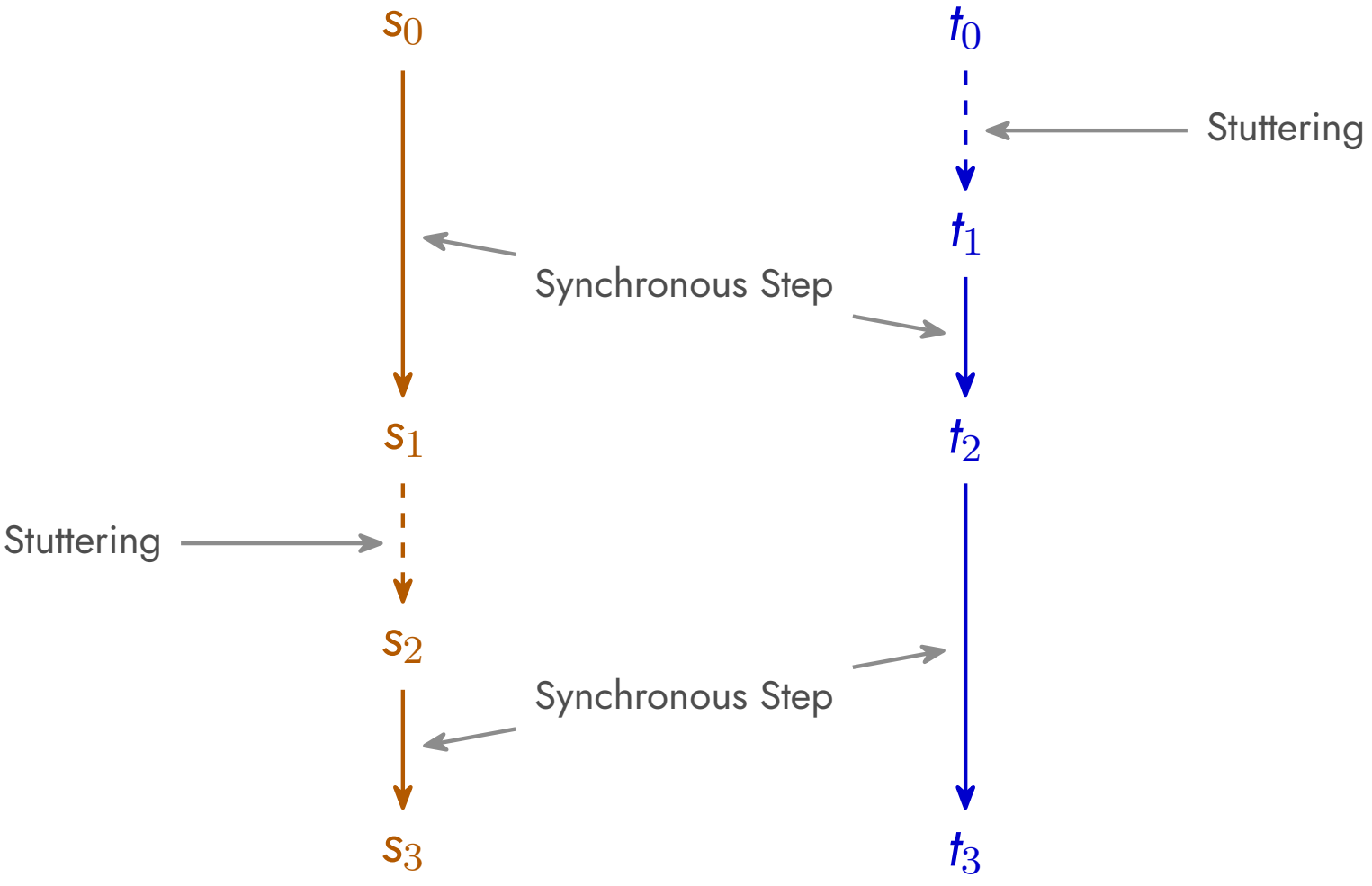

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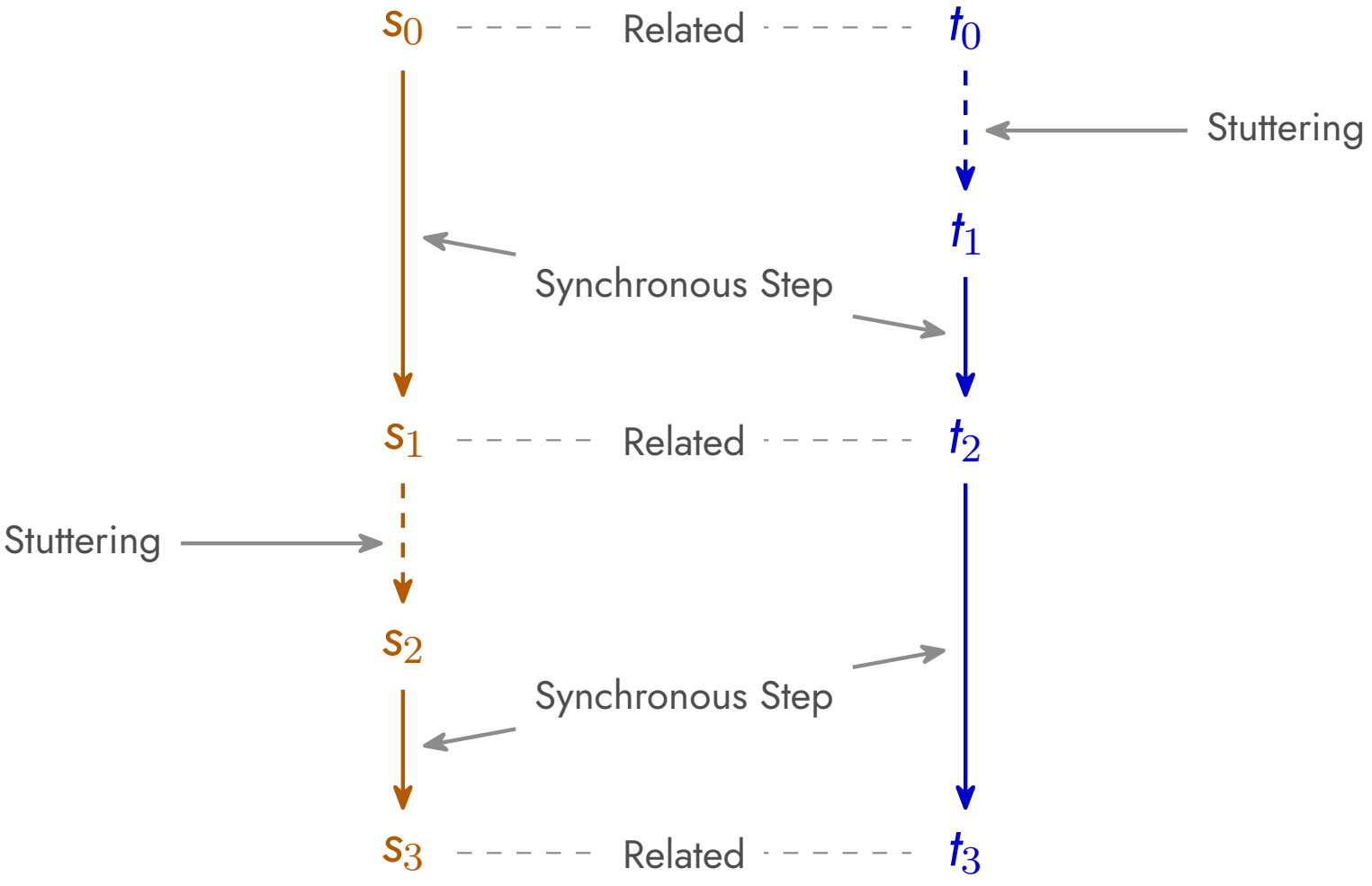

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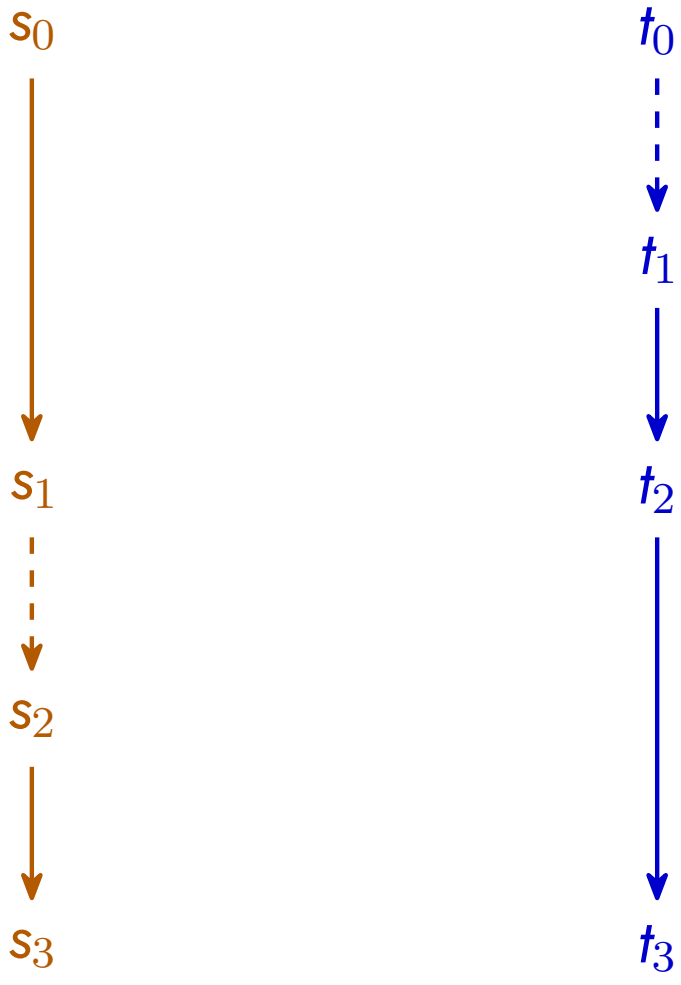

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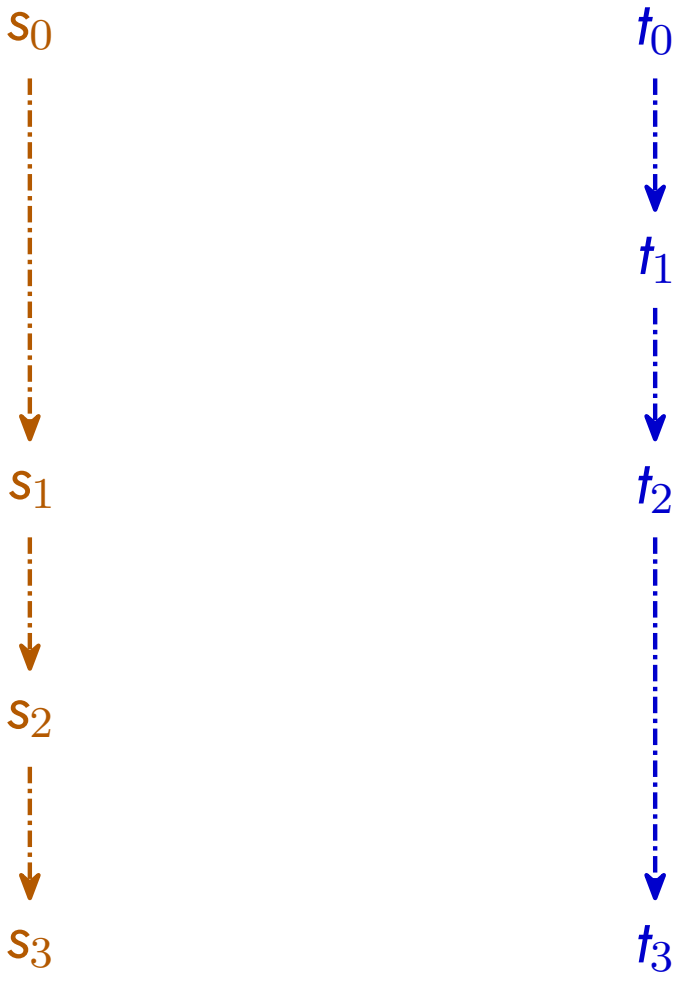

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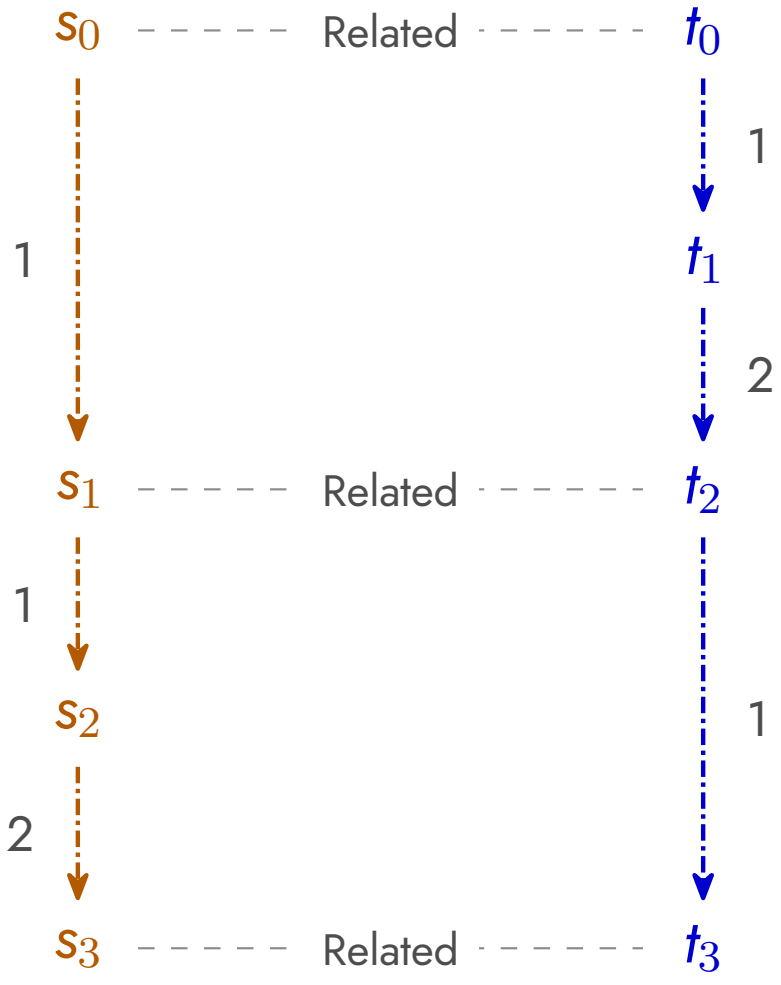

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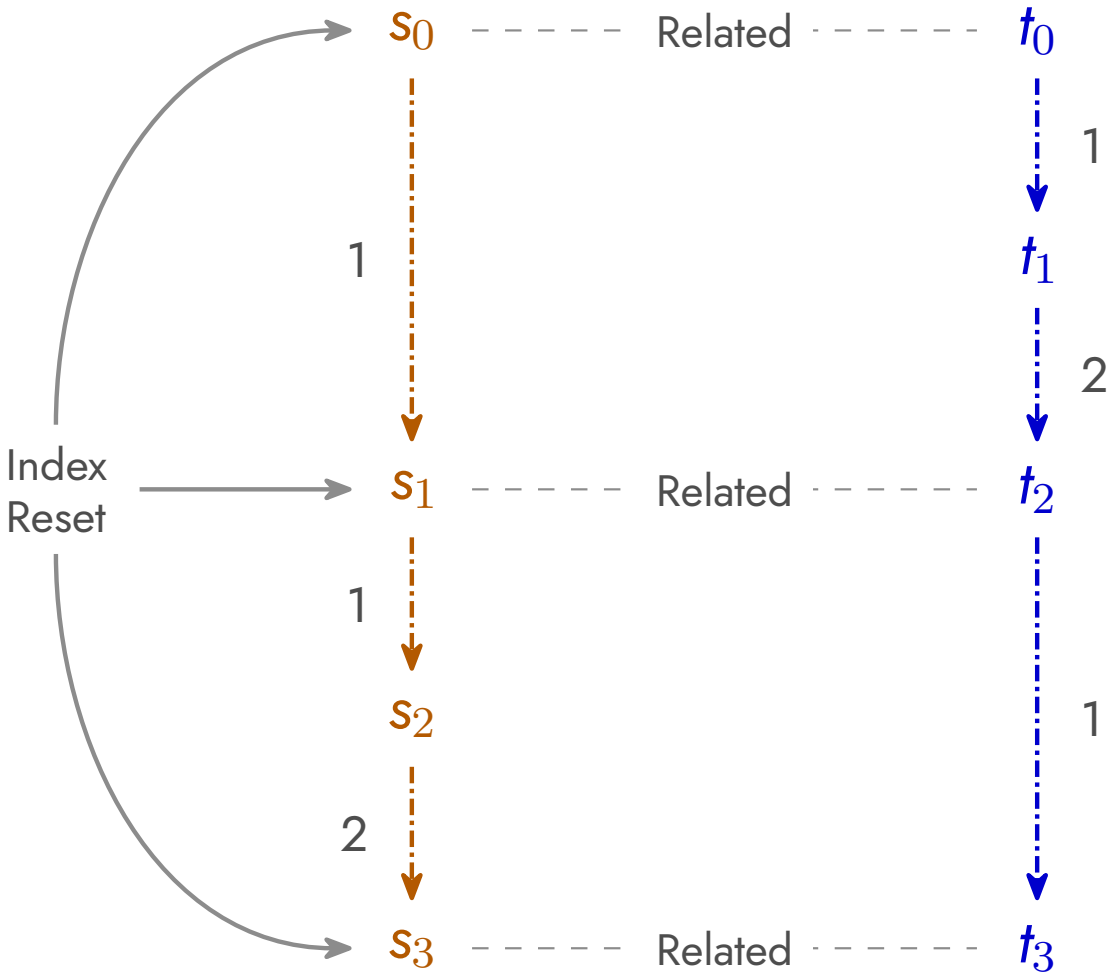

Theoretical Guarantees

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Soundness:

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Index Irrelevance:

$$(\exists i\, j,\ \langle T, i\rangle \sim_{\text{free}} \langle S, j\rangle) \implies \forall i\, j, \quad \langle T, i\rangle \sim_{\text{free}} \langle S, j\rangle$$

Asymmetric Reasoning

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Target	Source

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😈 nondeterminism	

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Target	Source
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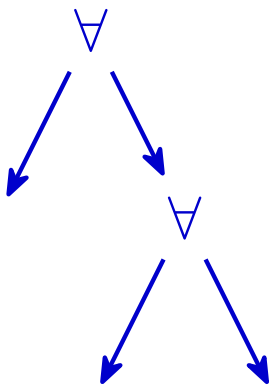

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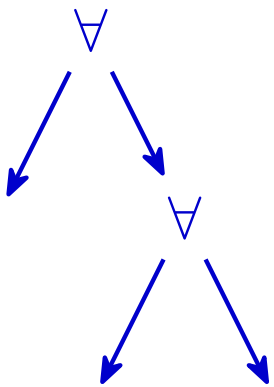
Target	Source
😈 nondeterminism	😇 nondeterminism



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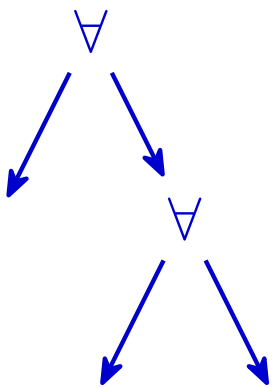

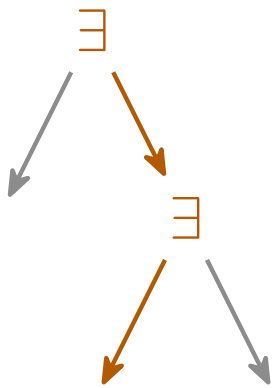

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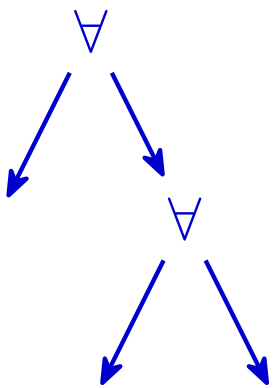
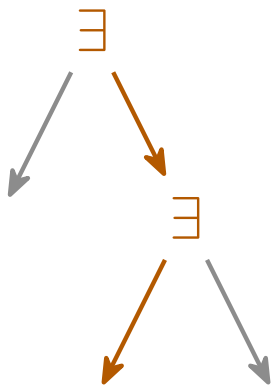
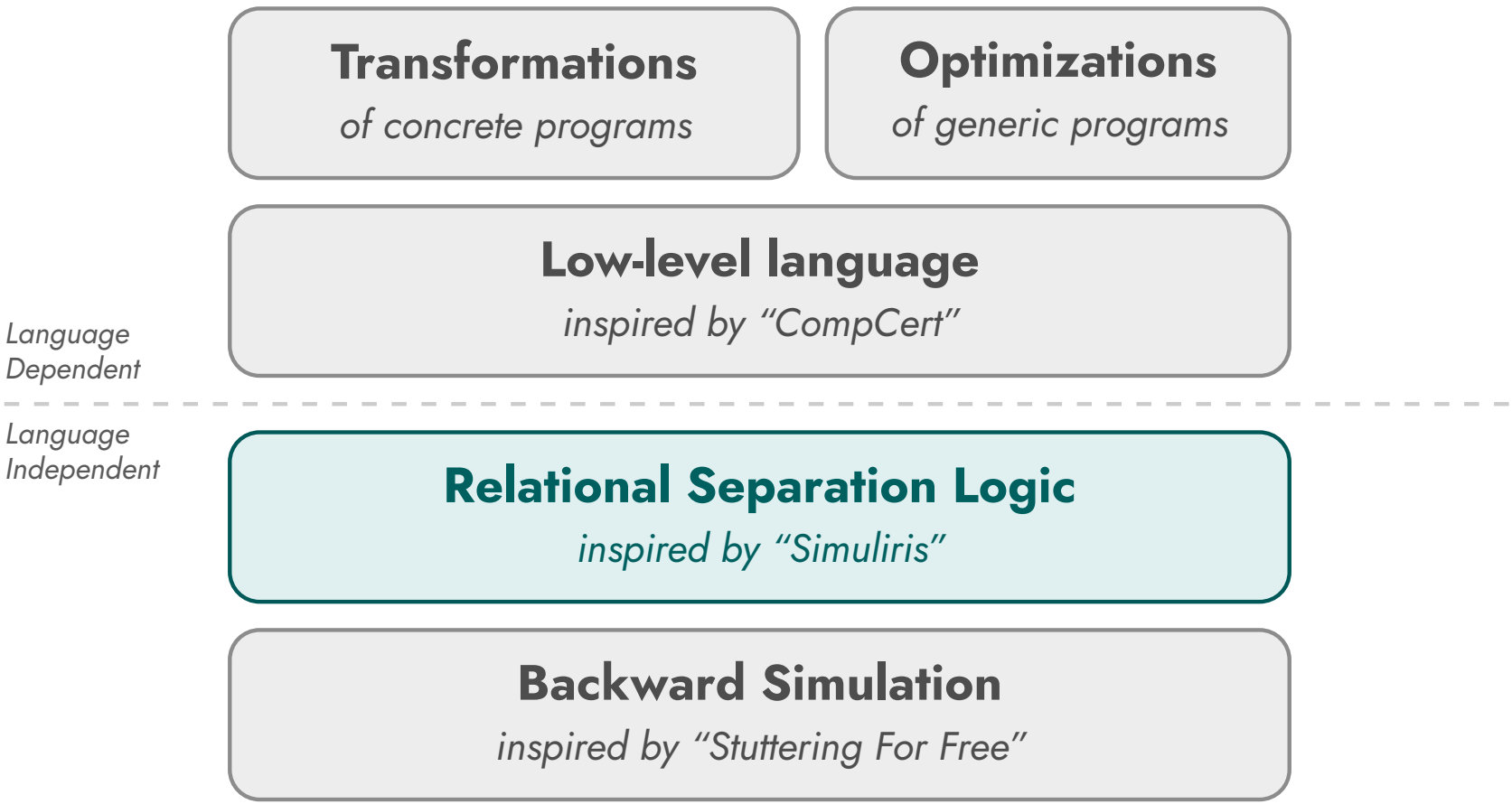




$$\lightning \in S \implies T \sim_{\text{free}} S$$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$$\triangleq P \twoheadrightarrow \mathsf{FreeSim}\,(f_t, f_s, Q)$$

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If $P\,(\vec{v_t}, \vec{v_s})$:

Relational Hoare Quadruples

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Relational Hoare Quadruples

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$$\triangleq P \twoheadrightarrow \mathsf{FreeSim}\,(f_t, f_s, Q)$$

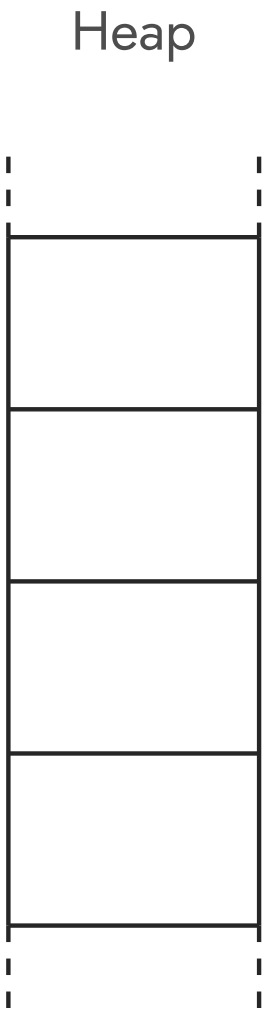
If $P\,(\vec{v_t}, \vec{v_s})$:

$$\lightning \in f_s$$

or

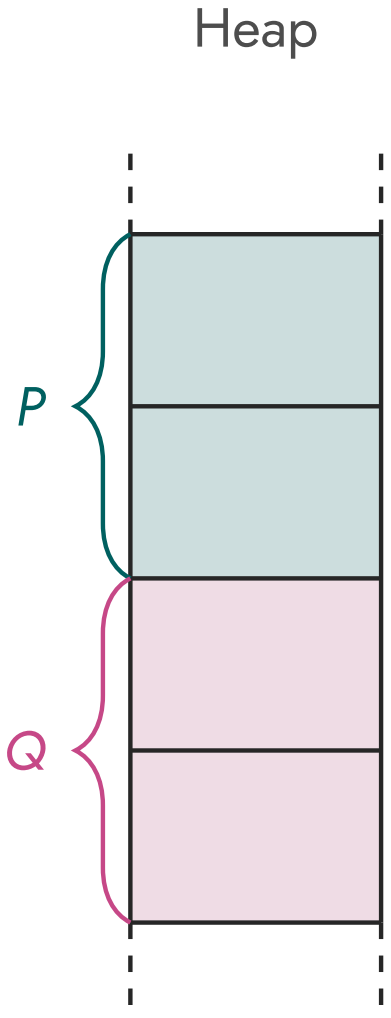
$$\begin{cases} \mathtt{Term}(v_t, m_t) \in f_t \implies \mathtt{Term}(v_s, m_s) \in f_s \text{ with } Q\,(v_t, v_s, m_t, m_s) \\ \mathtt{Div} \in f_t \implies \mathtt{Div} \in f_s \end{cases}$$

Relational Separation Logic



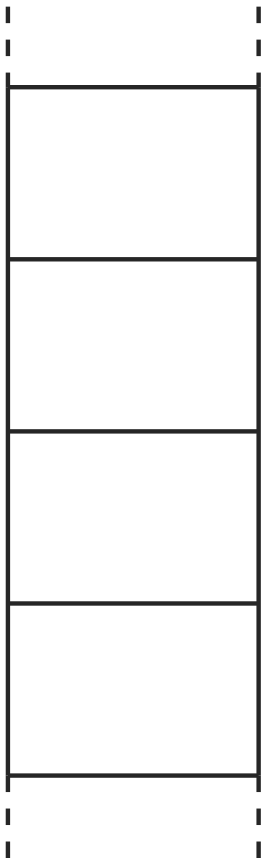
Relational Separation Logic

$P * Q \implies$

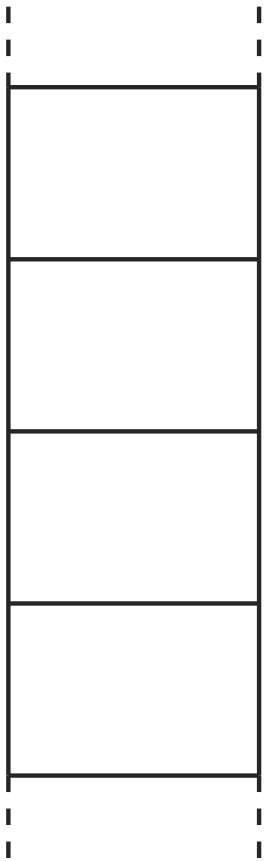


Relational Separation Logic

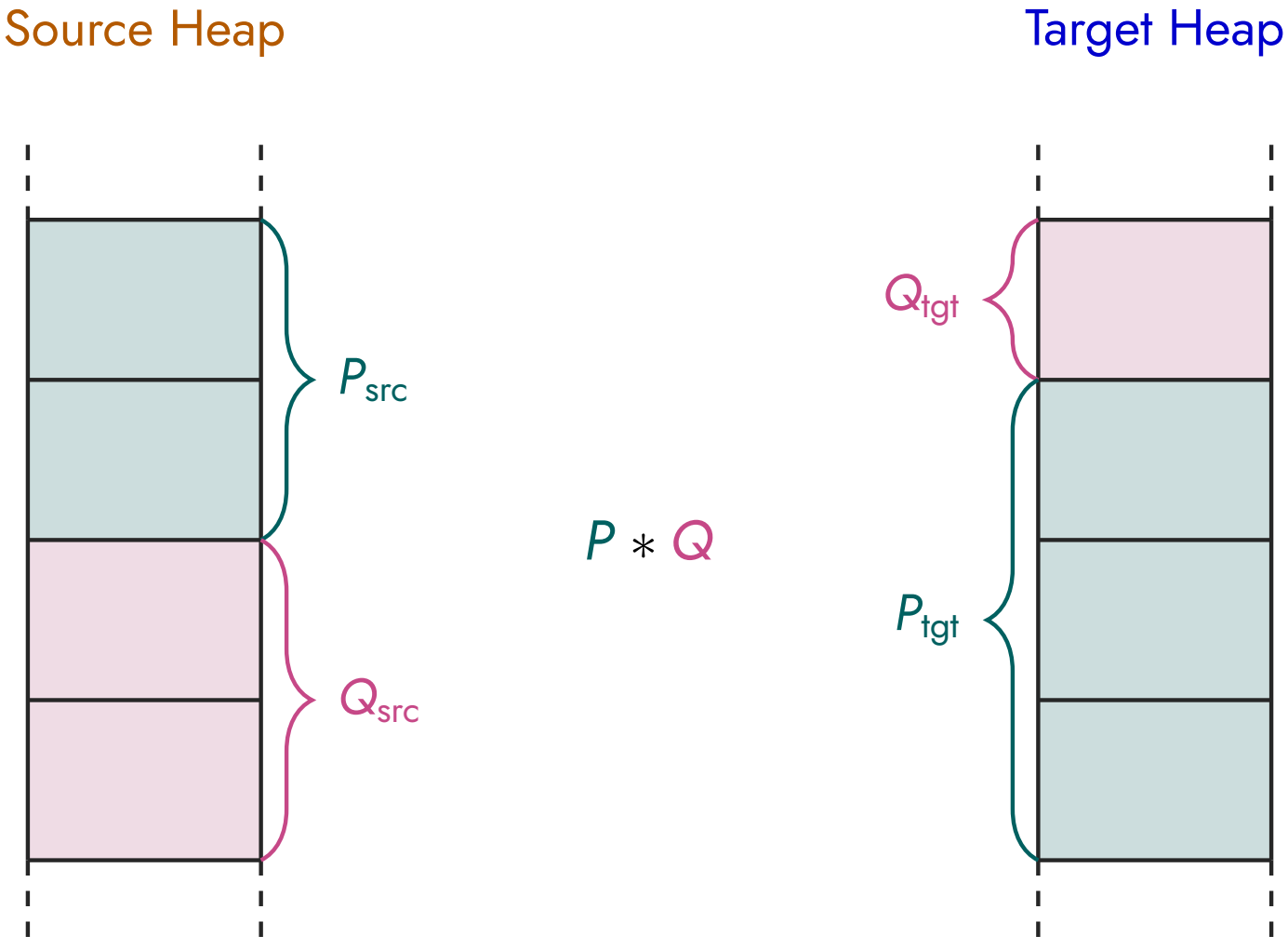
Source Heap



Target Heap



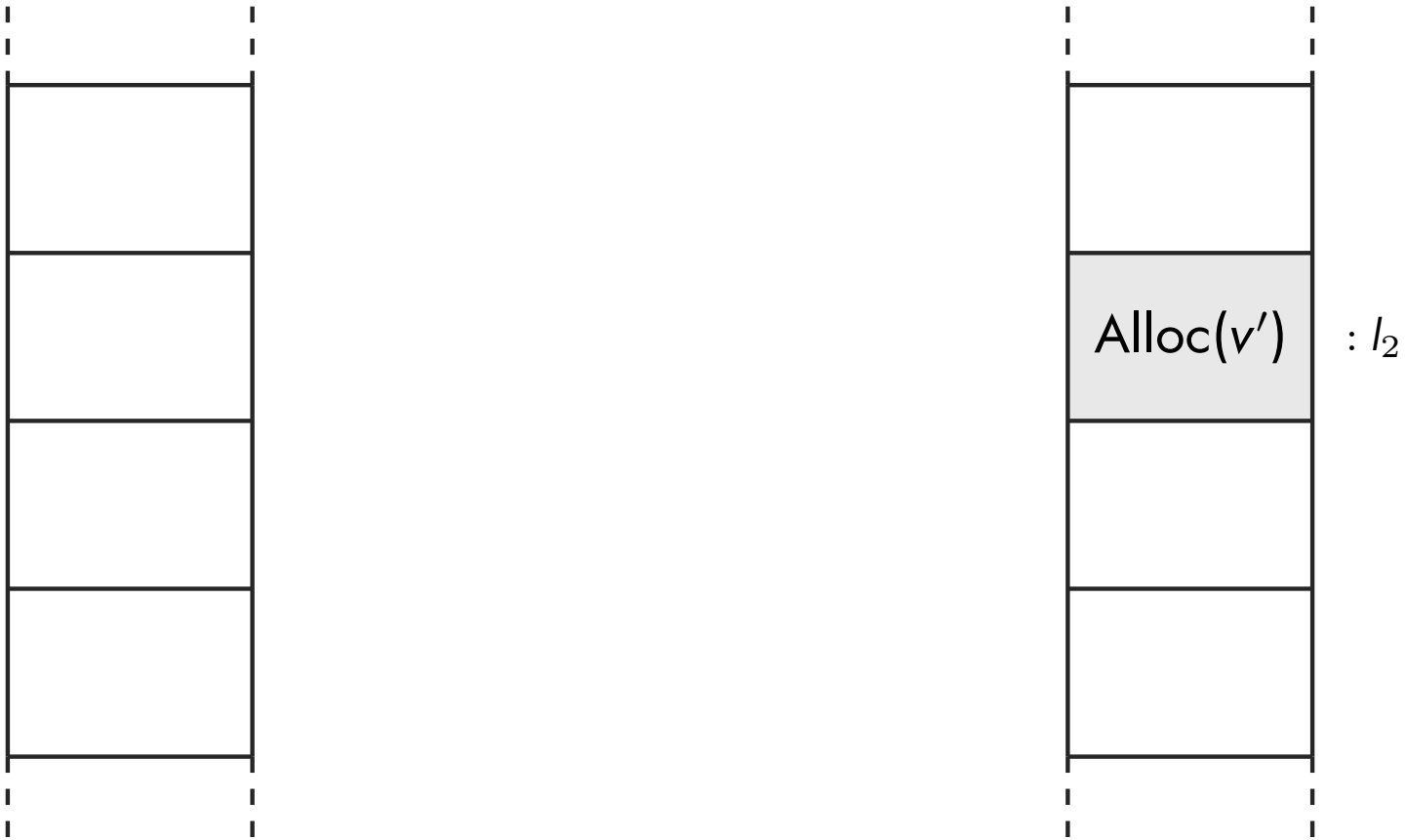
Relational Separation Logic



Relational Separation Logic

Source Heap

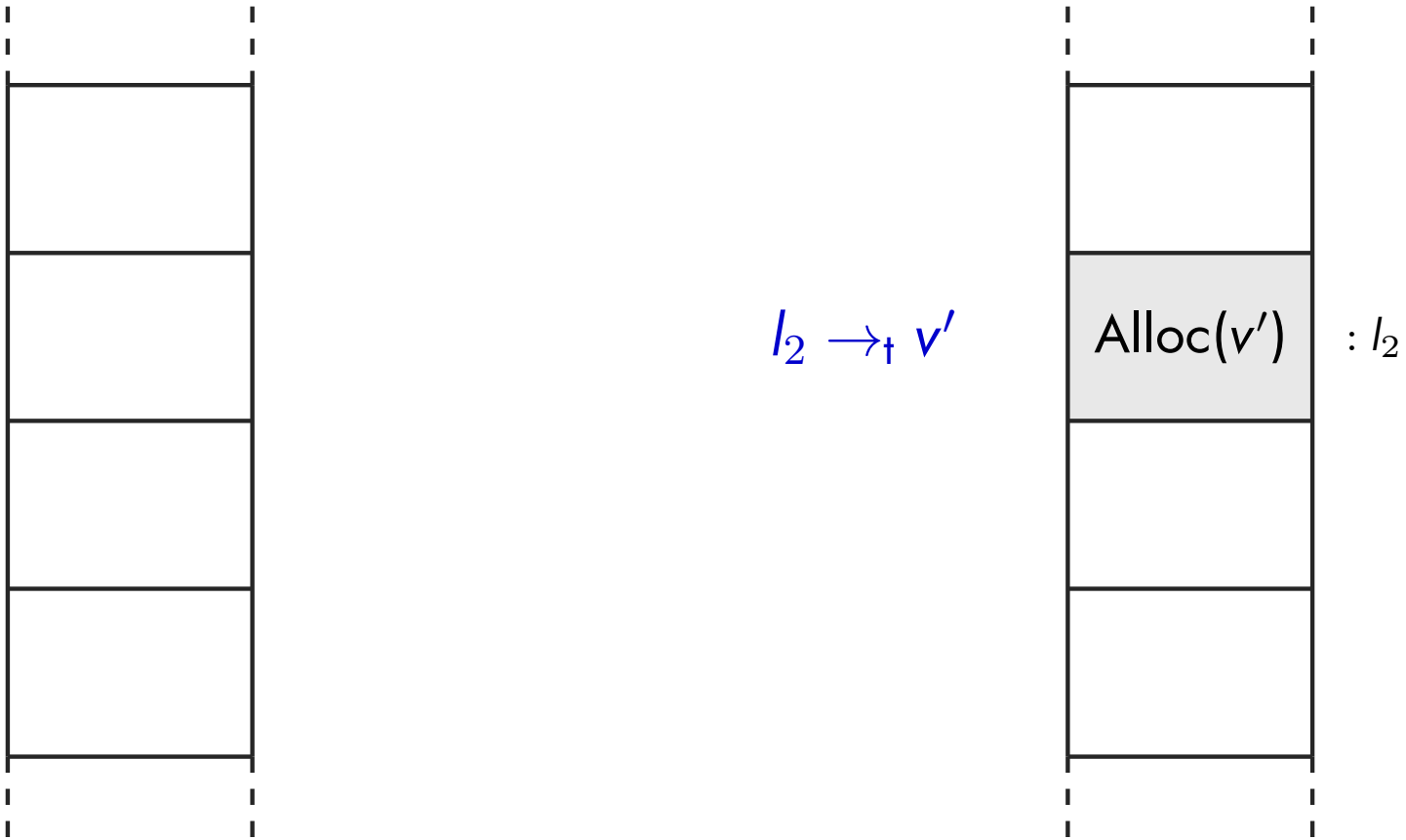
Target Heap



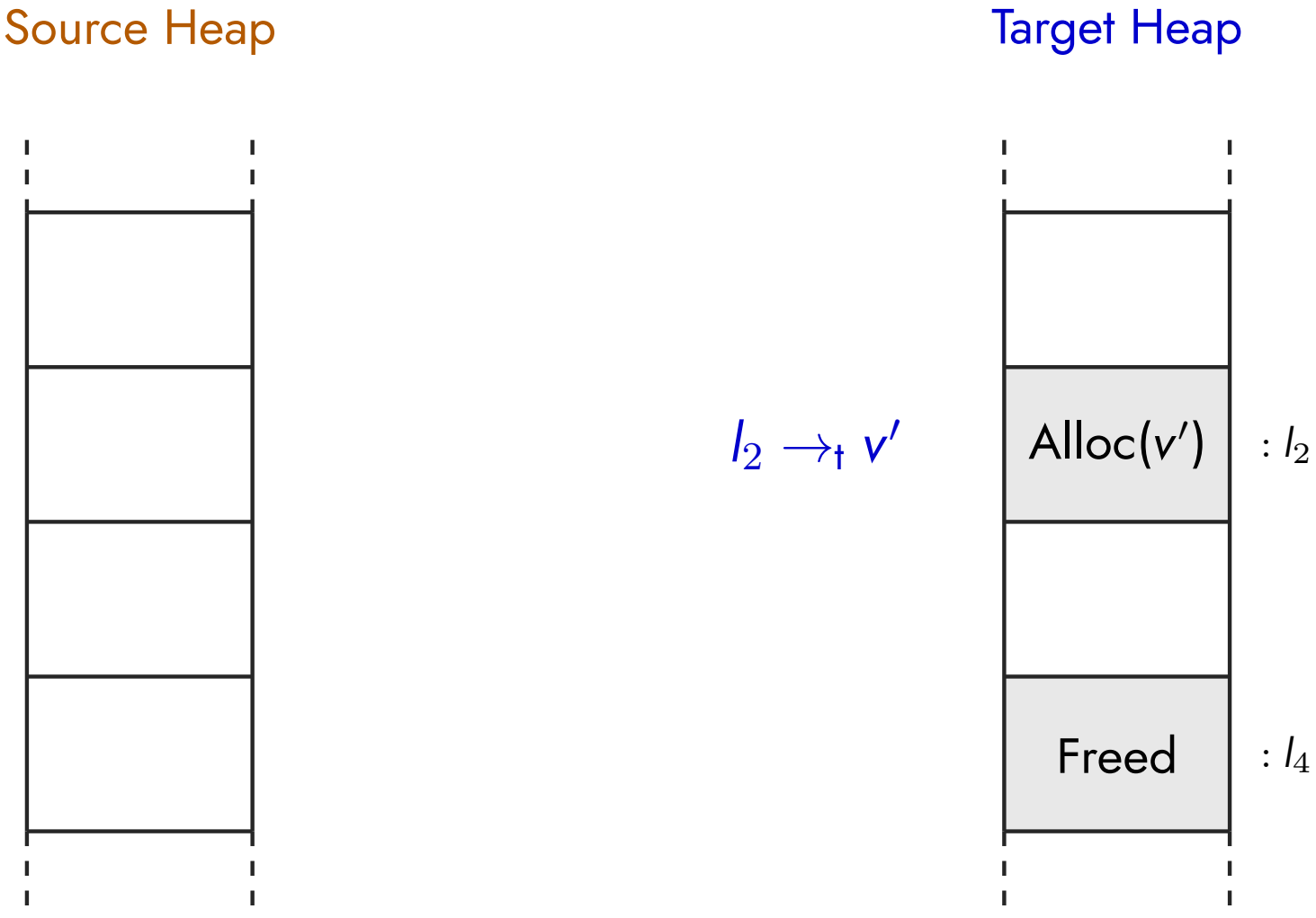
Relational Separation Logic

Source Heap

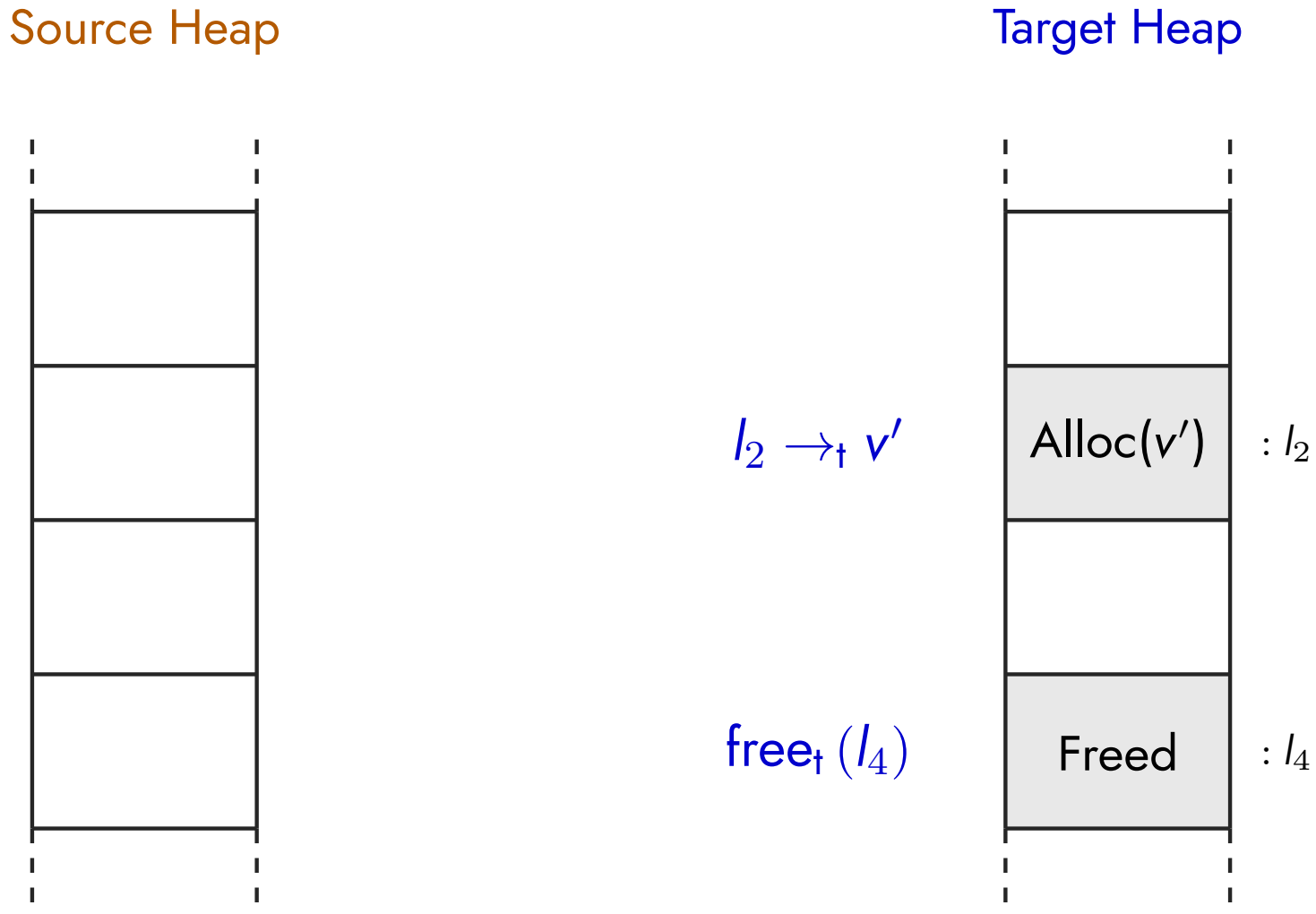
Target Heap



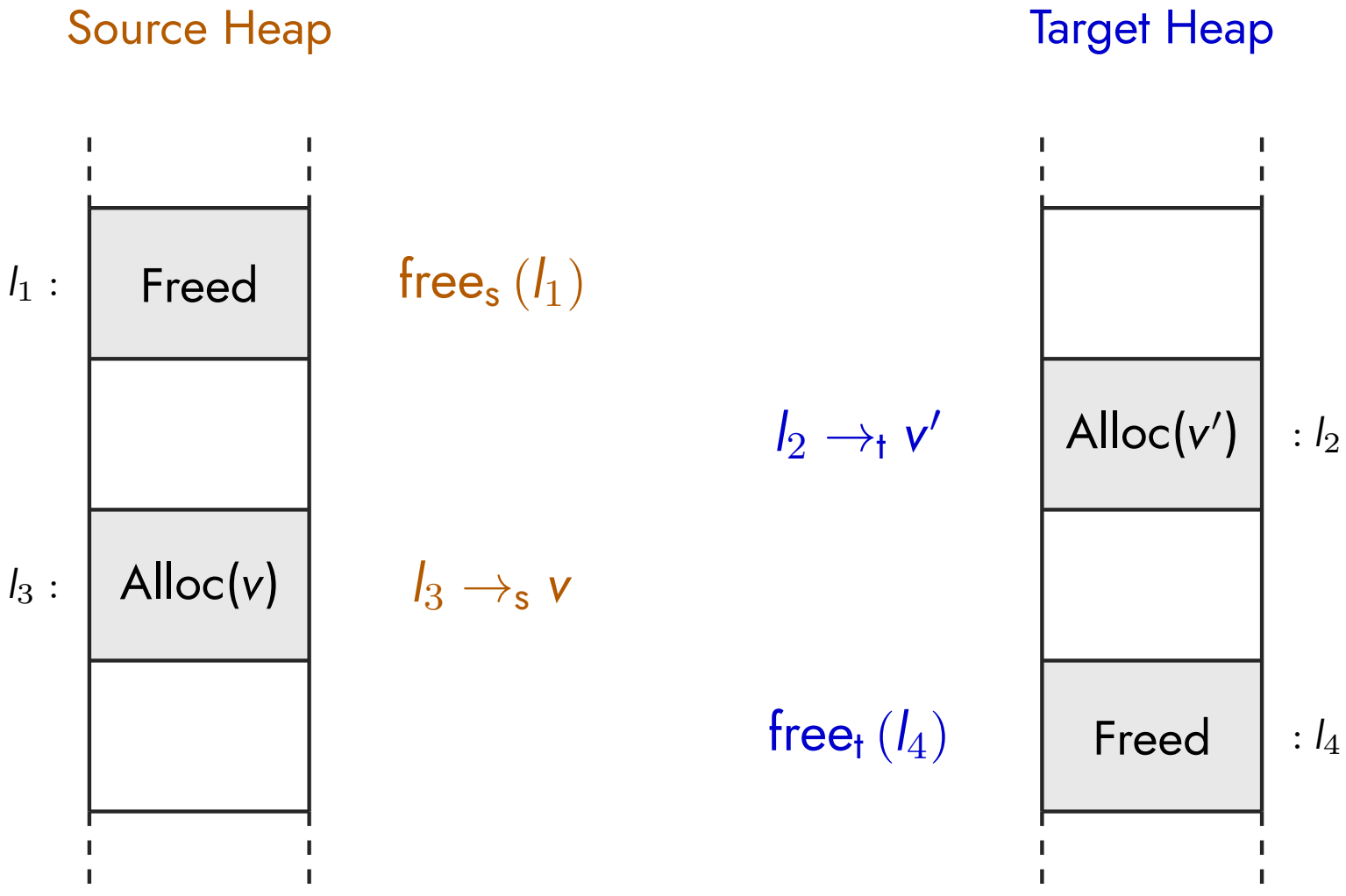
Relational Separation Logic



Relational Separation Logic

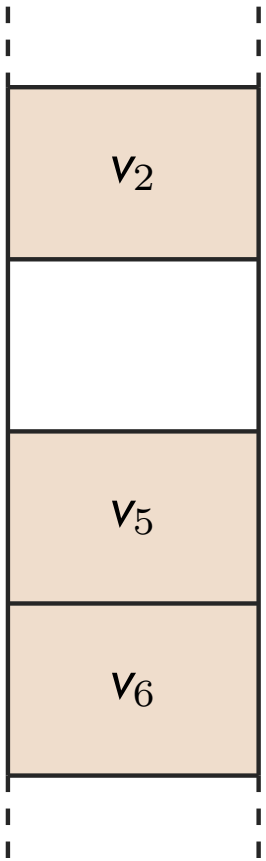


Relational Separation Logic

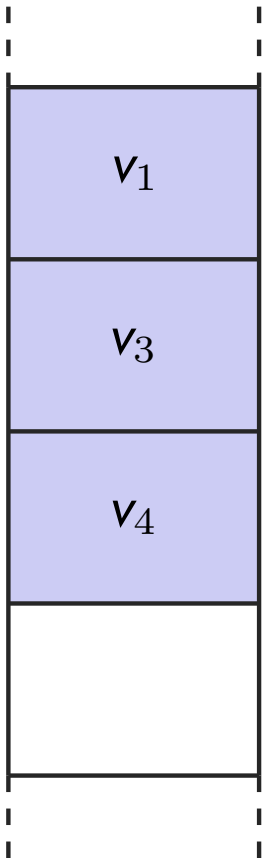


Value Equivalence

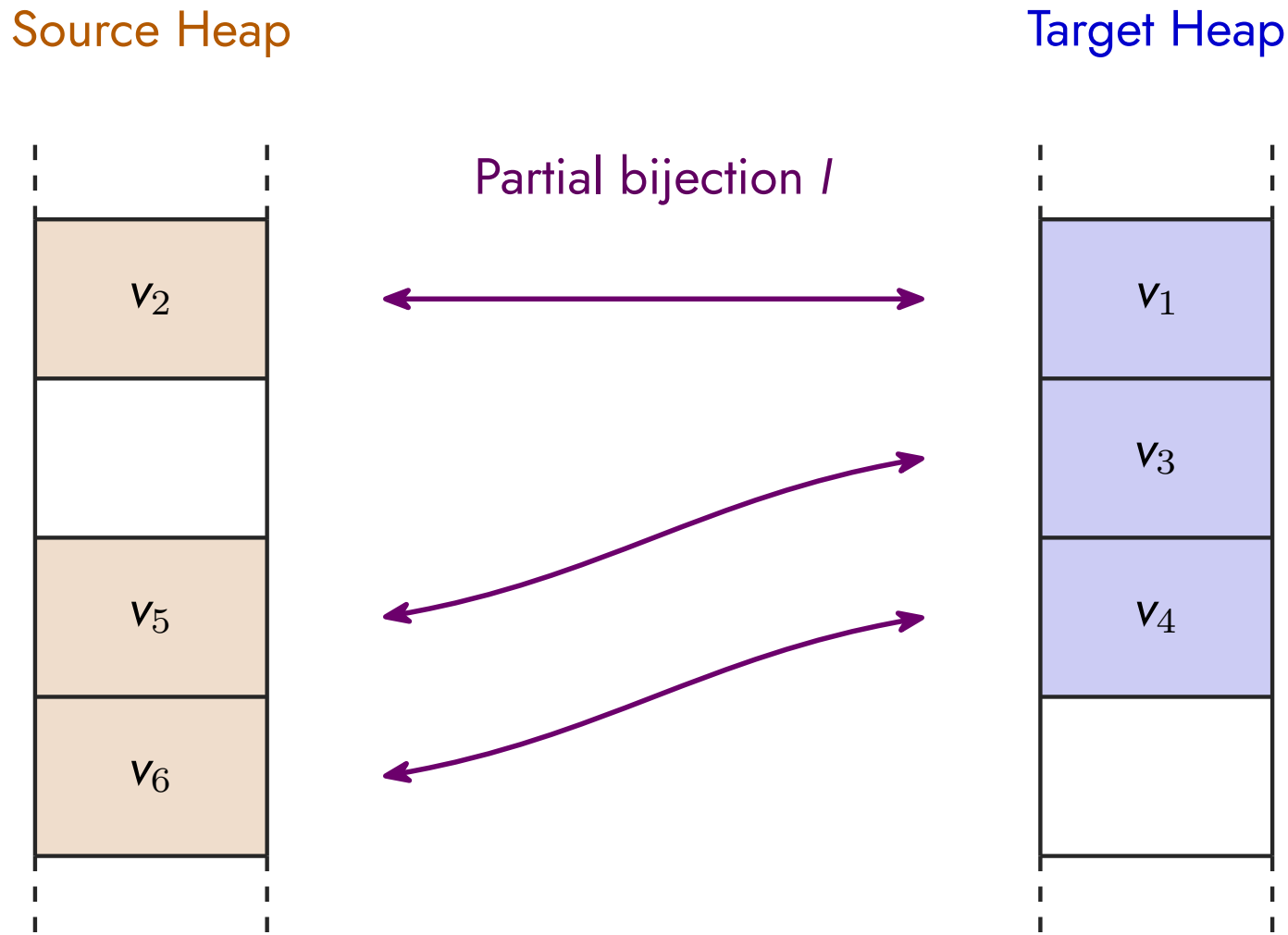
Source Heap



Target Heap



Value Equivalence



Value Equivalence

$$V_t \bowtie_I V_s$$

Value Equivalence

$$V_t \bowtie_I V_s$$

$$i_t = i_s \quad \Longrightarrow \quad \mathsf{VInt}\,(i_t) \bowtie_I \mathsf{VInt}\,(i_s)$$

Value Equivalence

$$v_t \bowtie_l v_s$$

$$i_t = i_s \implies \mathsf{VInt}\,(i_t) \bowtie_l \mathsf{VInt}\,(i_s)$$

$$b_t = b_s \implies \mathsf{VBool}\,(b_t) \bowtie_l \mathsf{VBool}\,(b_s)$$

Value Equivalence

$$v_t \bowtie_I v_s$$

$$i_t = i_s \implies \mathsf{VInt}\,(i_t) \bowtie_I \mathsf{VInt}\,(i_s)$$

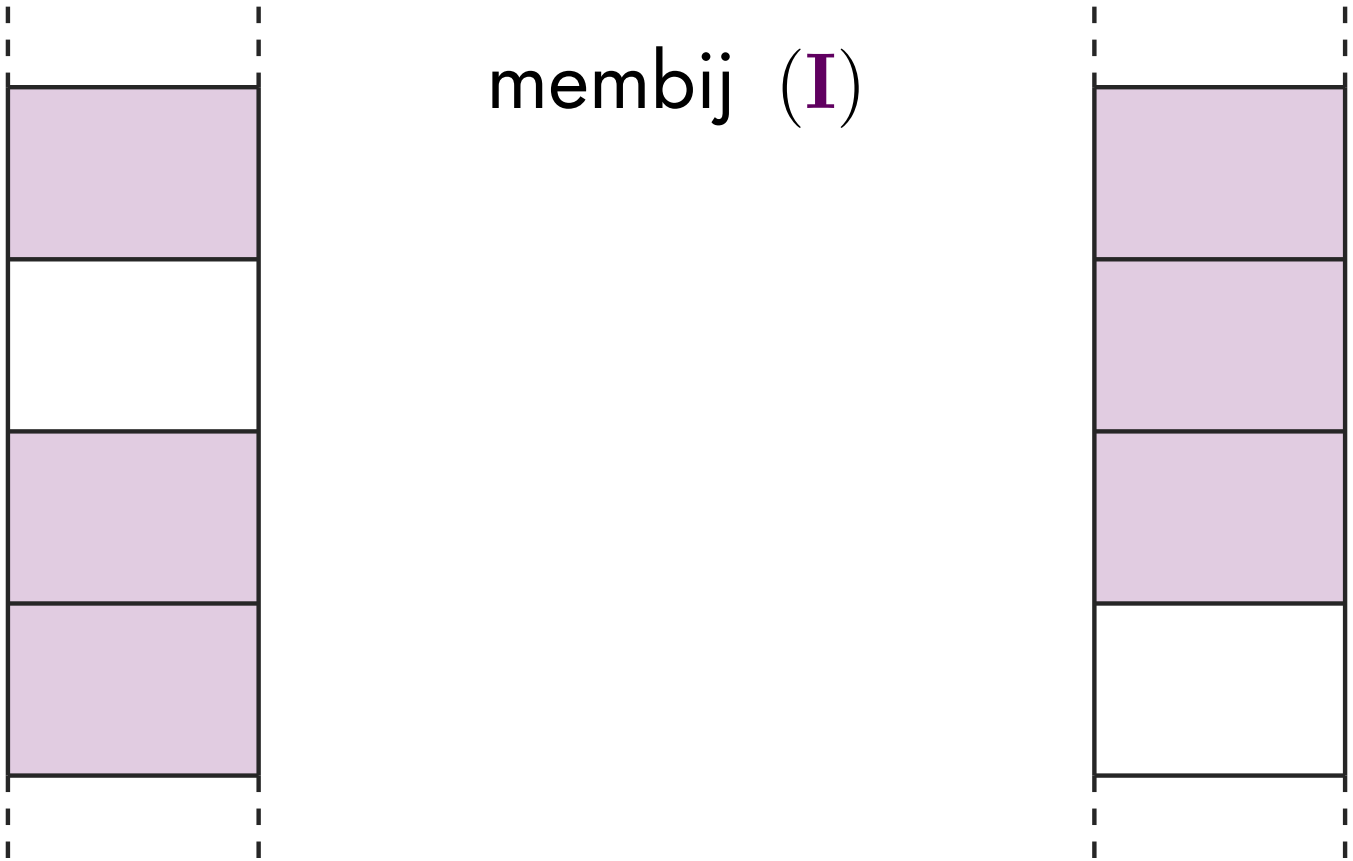
$$b_t = b_s \implies \mathsf{VBool}\,(b_t) \bowtie_I \mathsf{VBool}\,(b_s)$$

$$(l_t, l_s) \in I \implies \mathsf{VPtr}\,(l_t) \bowtie_I \mathsf{VPtr}\,(l_s)$$

Memory Bijection

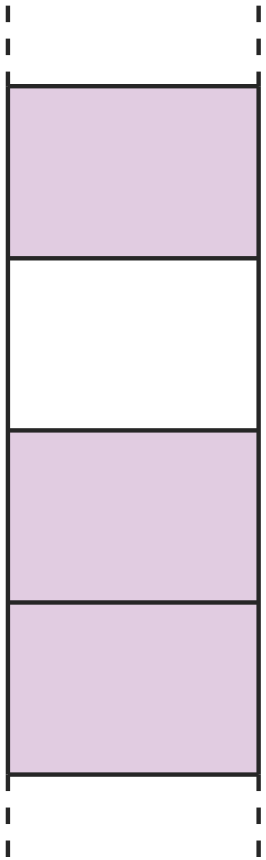
Source Heap

Target Heap



Memory Bijection

Source Heap



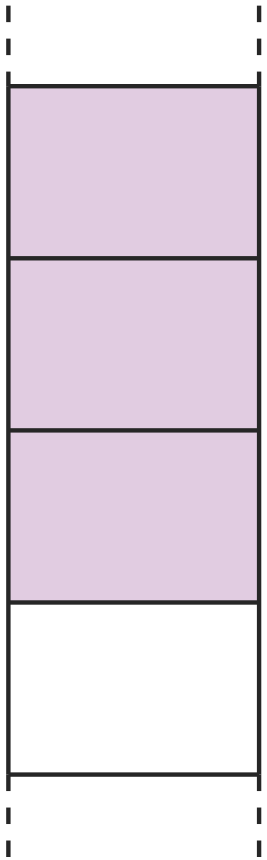
membij $(\mathbf{I})$

$(l_t, l_s) \in I$

$\Downarrow$

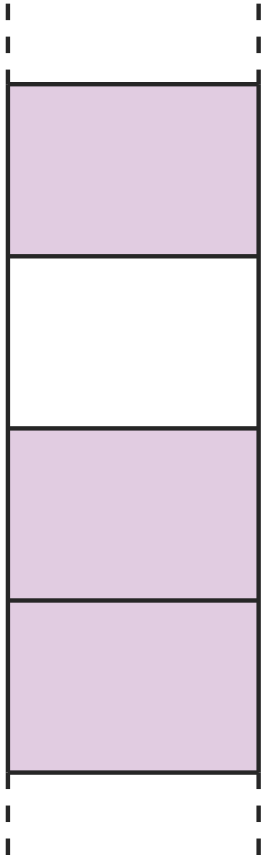
both alive with $\bowtie_I$

Target Heap

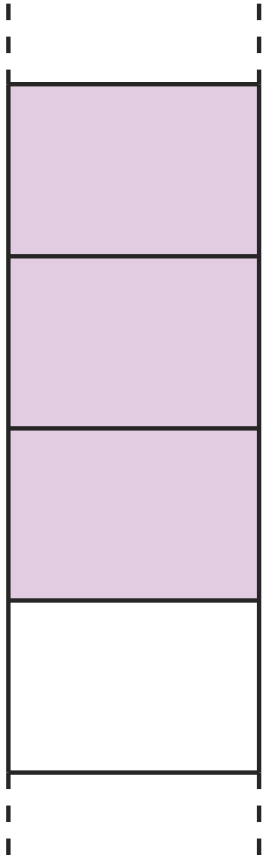


Memory Bijection

Source Heap



Target Heap



membij $(\mathbf{I})$

$(l_t, l_s) \in I$

$\Downarrow$

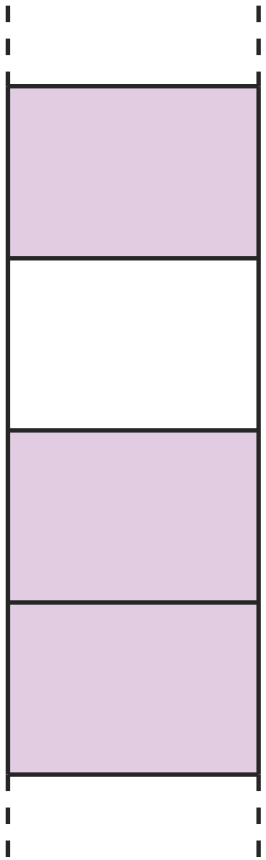
both alive with $\bowtie_I$

or

both freed

Memory Bijection

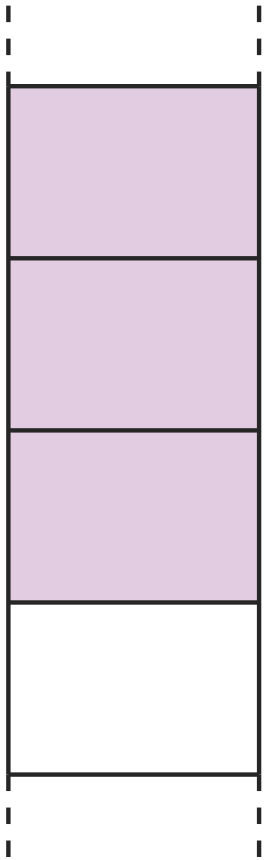
Source Heap



$\text{membij}_{\mathbf{E}}(\mathbf{I})$

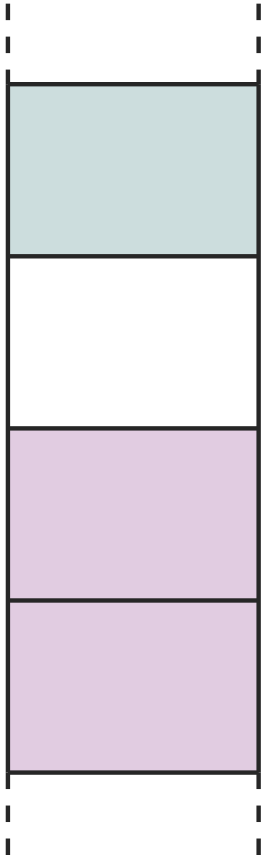
Exceptions

Target Heap



Memory Bijection

Source Heap

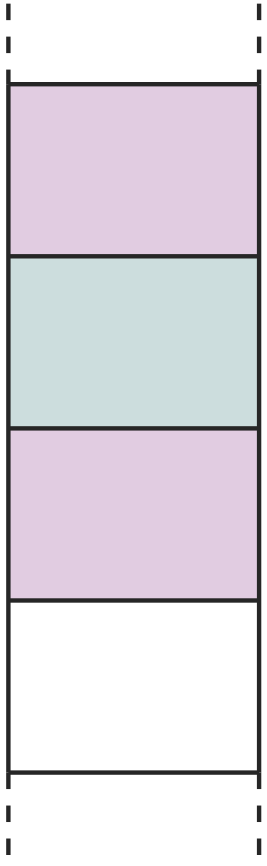


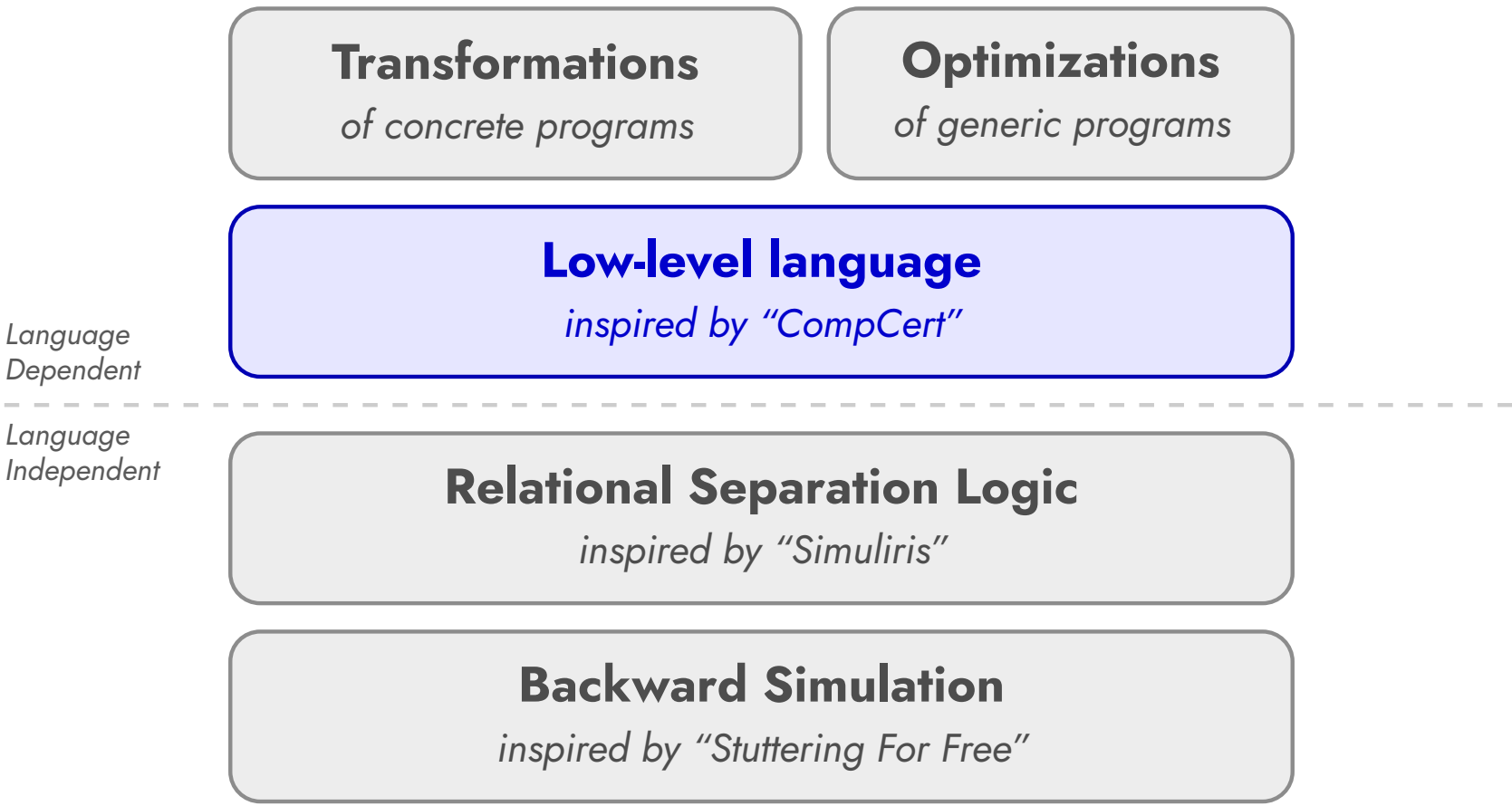
$\mathrm{membij}_{\mathbf{E}}(\mathbf{I})$

Exceptions

$E \subseteq I$

Target Heap

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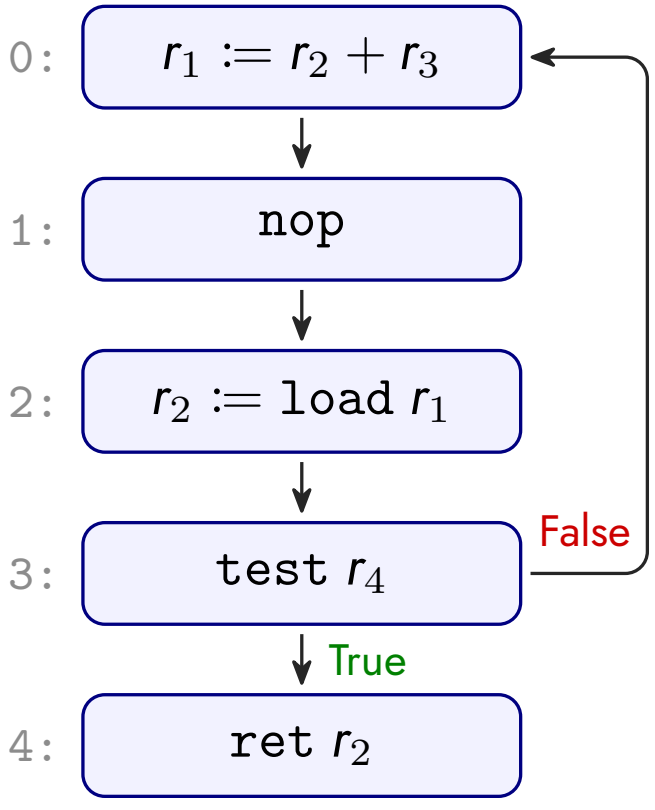
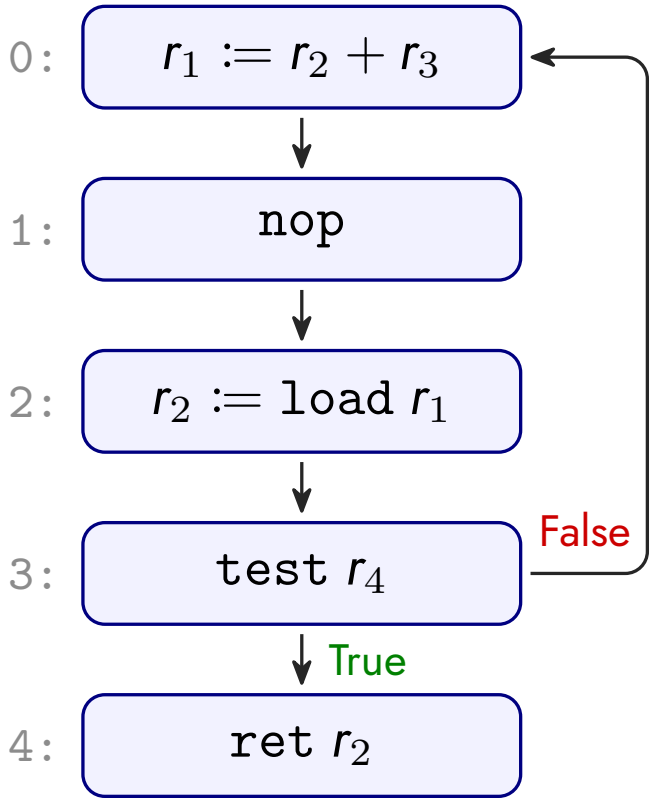
μRTL

μRTL



Function body

μRTL



Function body

Local Registers

r_1 r_2

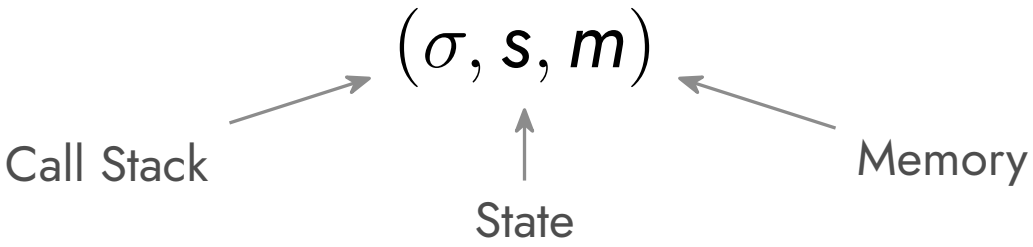
r_3 r_4

Operational Semantics

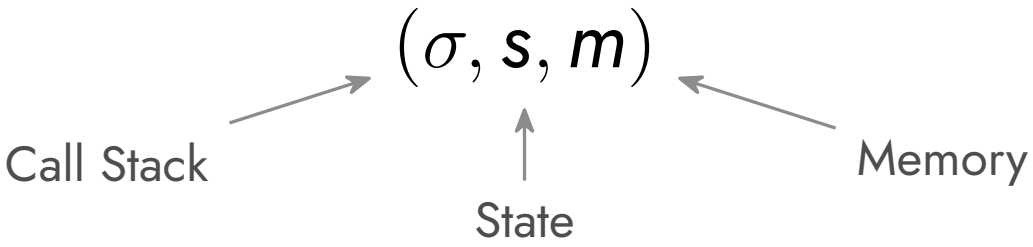
Operational Semantics

$$(\sigma, s, m)$$

Operational Semantics

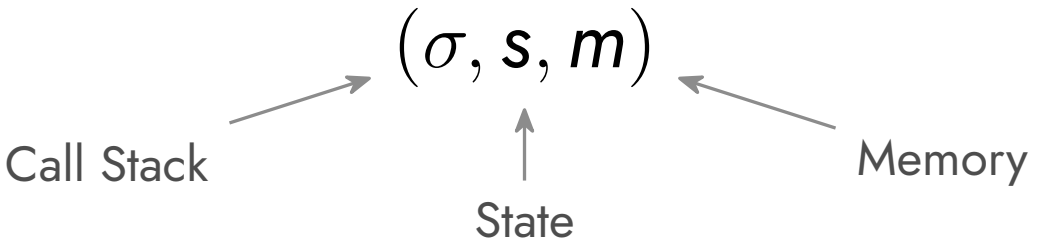


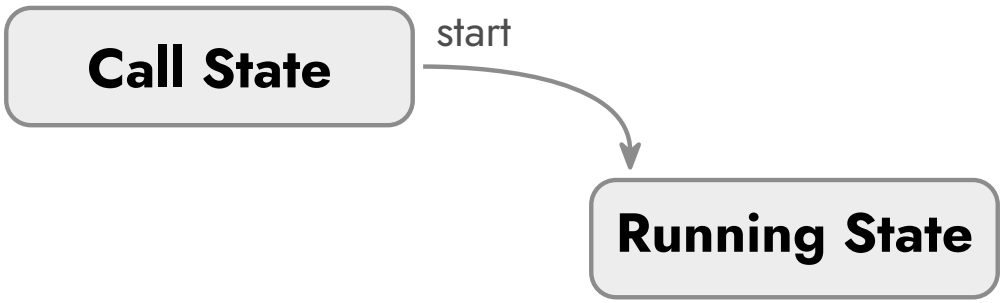
Operational Semantics



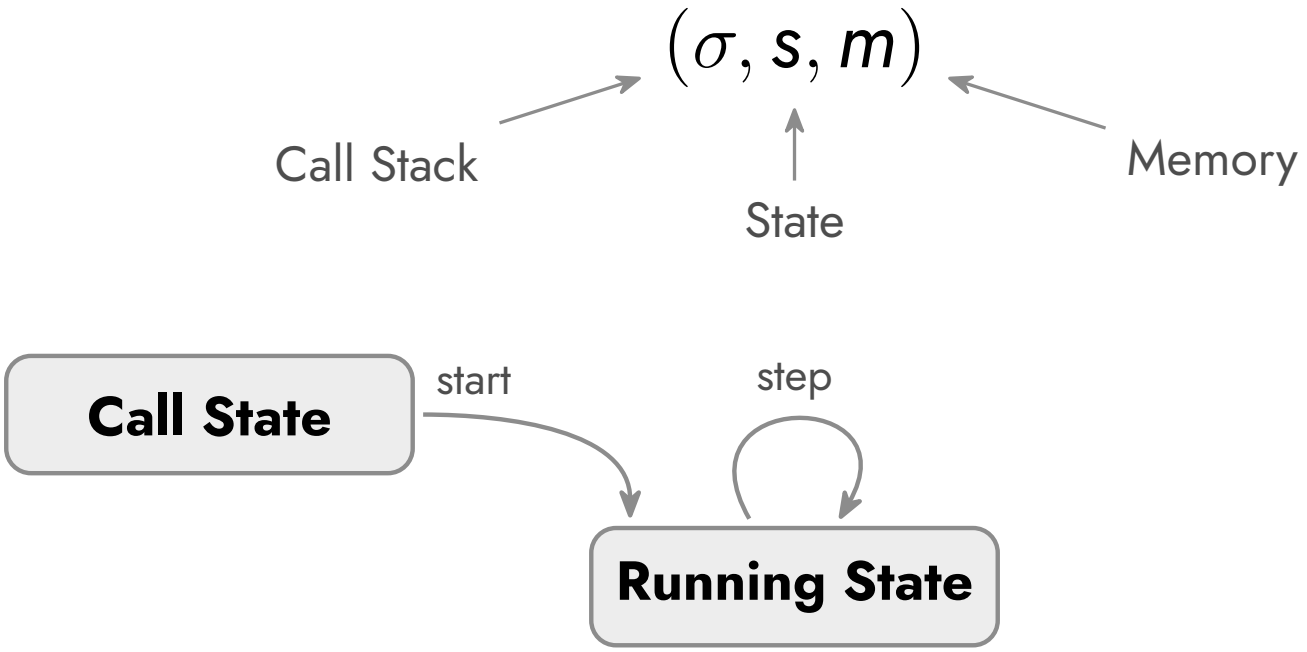
Call State

Operational Semantics

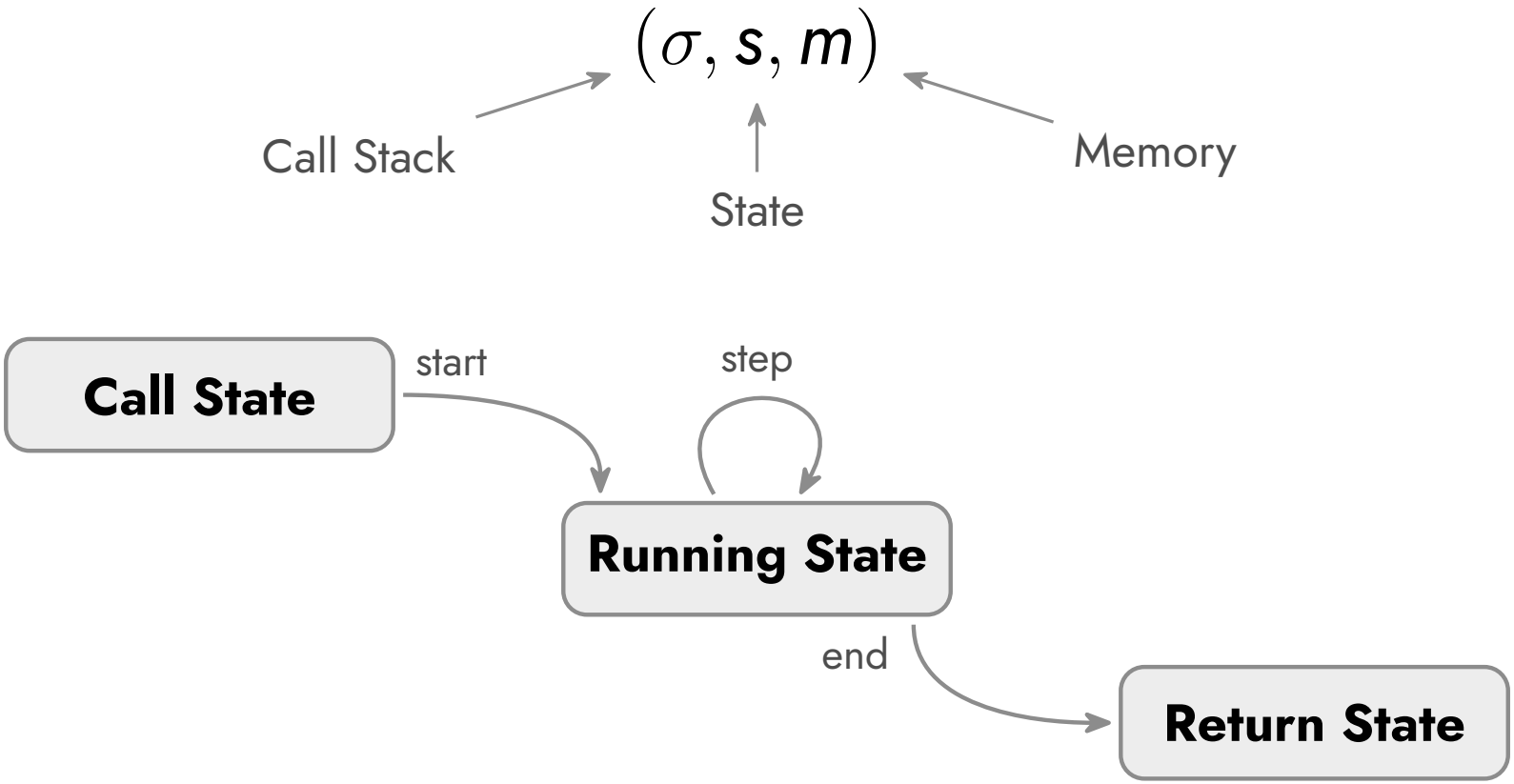




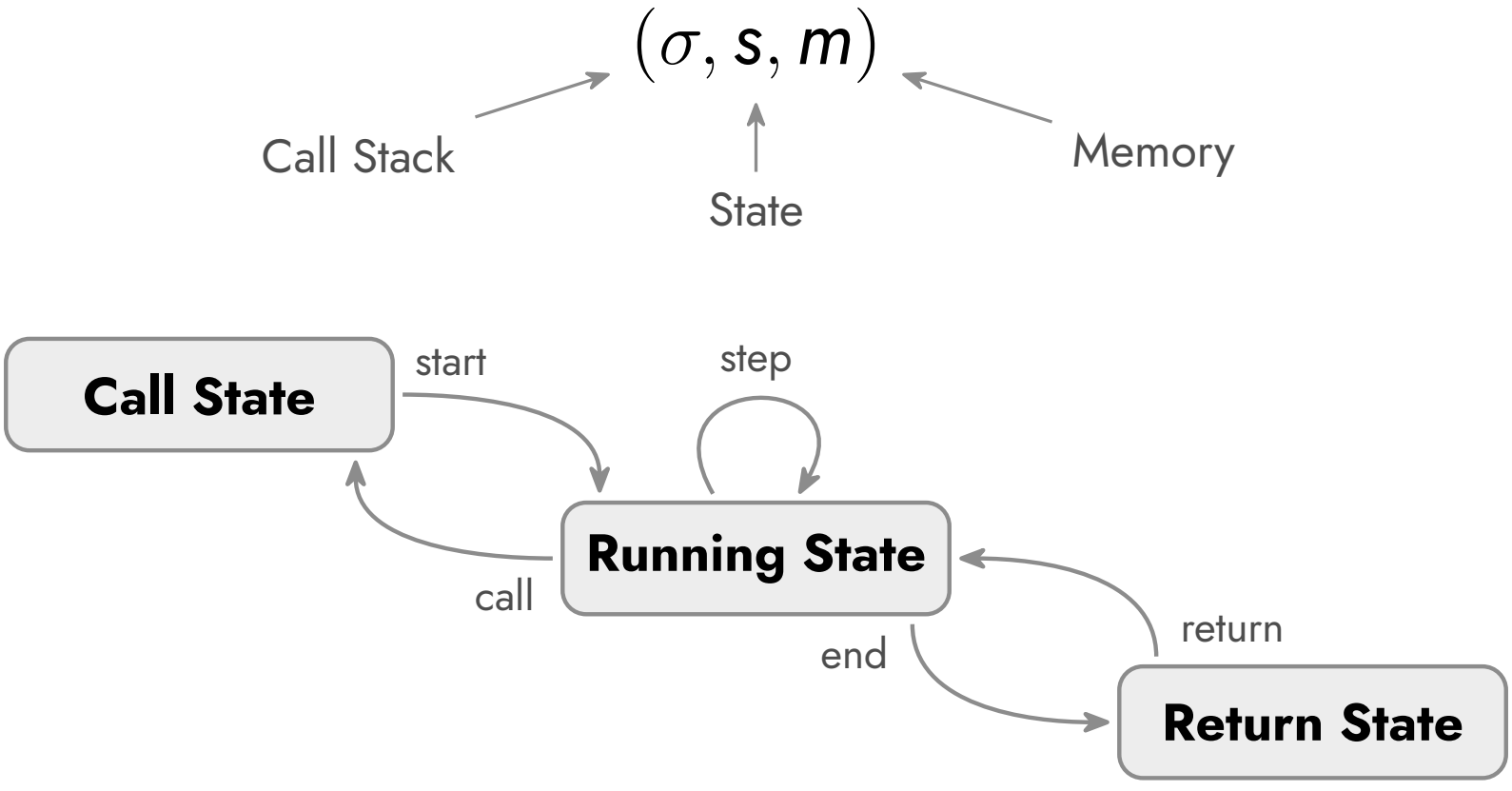
Operational Semantics



Operational Semantics



Operational Semantics



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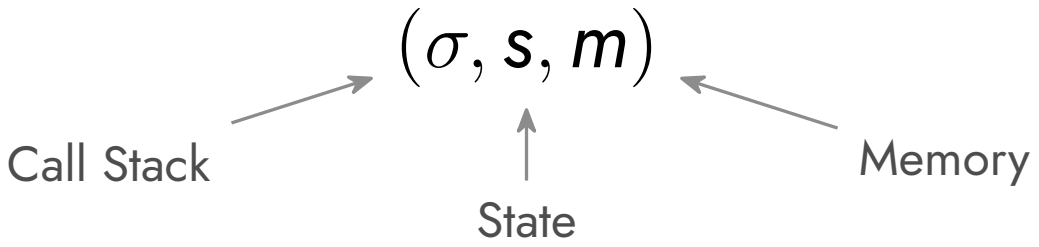
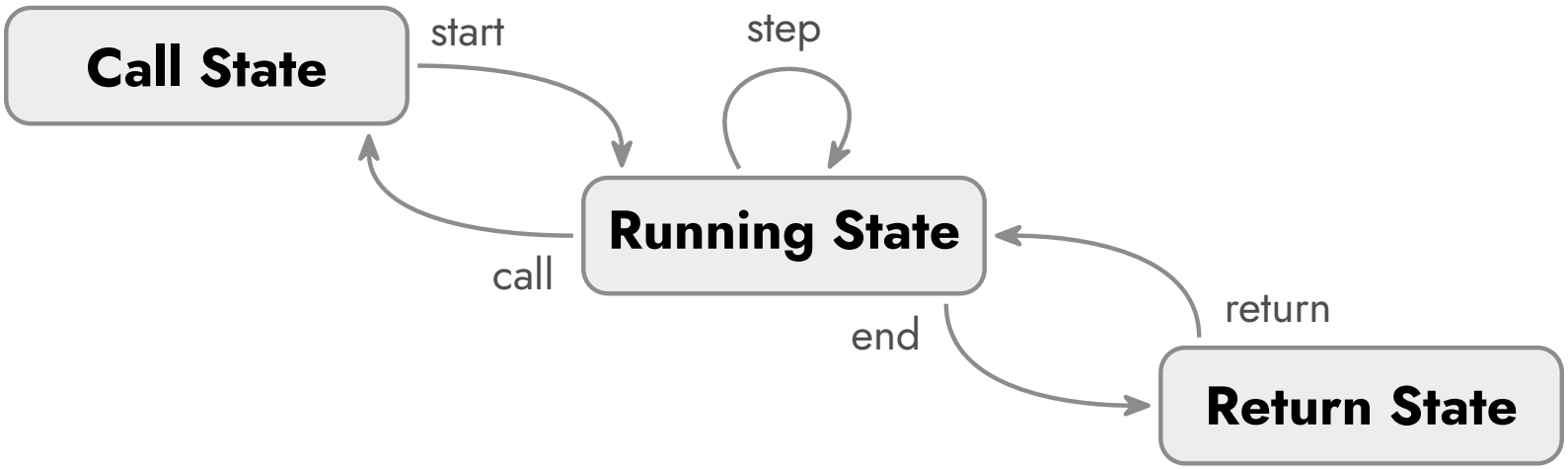
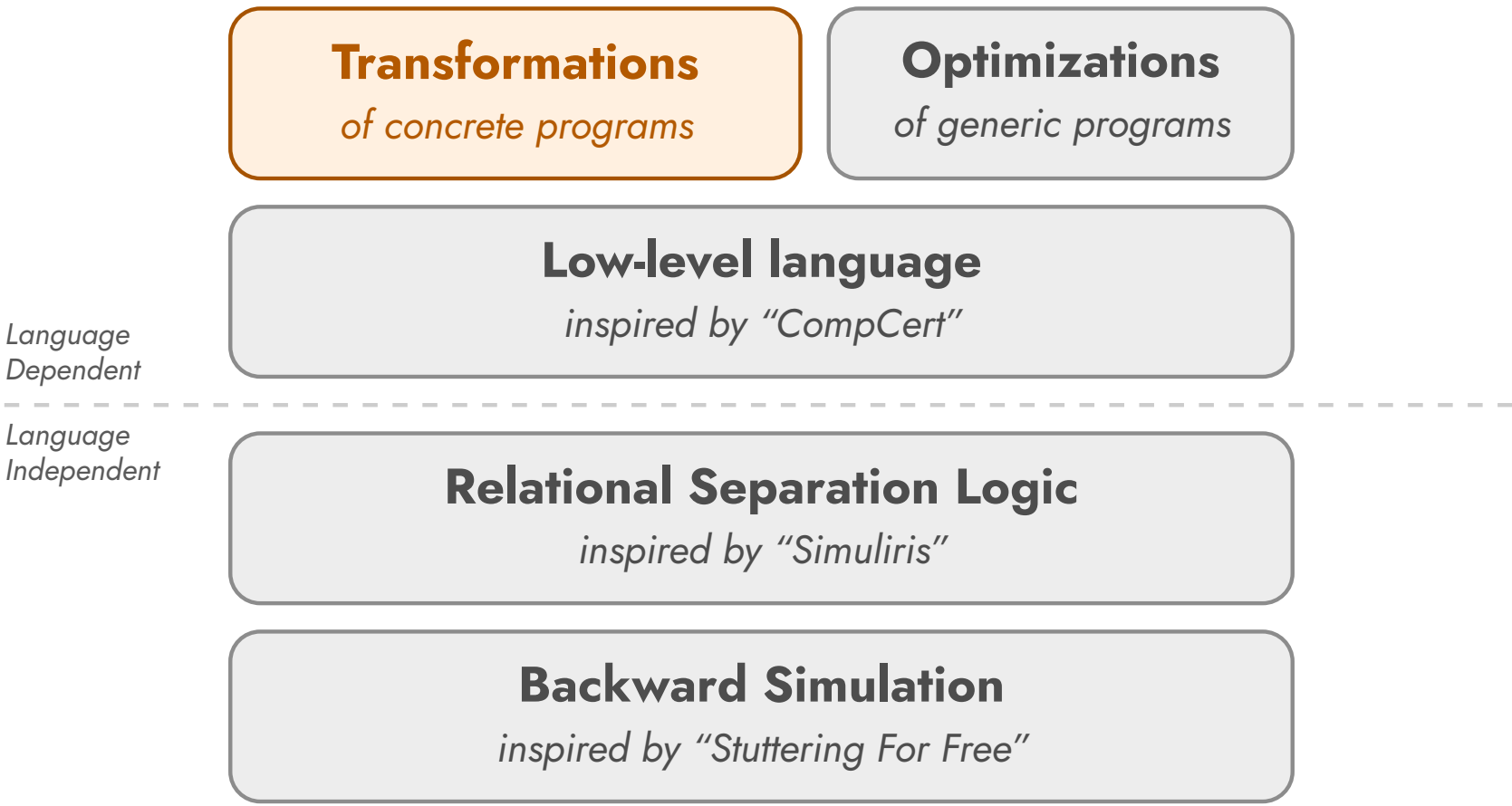




Modularity:

$$([], s, m) \twoheadrightarrow ([], s', m') \implies (\sigma, s, m) \twoheadrightarrow (\sigma, s', m')$$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Eliminating a Redundant Zero-Check

Eliminating a Redundant Zero-Check

Source Code

$1:\quad \ldots \rightarrow 2$

$2:\quad r_1 := r_n / r_d \rightarrow 3$

$3:\quad r_2 := \texttt{test}\ r_d \rightarrow 4$

$4:\quad \ldots$

Eliminating a Redundant Zero-Check

Source Code

1 : $\ldots \rightarrow 2$

2 : $r_1 := r_n / r_d \rightarrow 3$

3 : $r_2 := \texttt{test}\ r_d \rightarrow 4$

4 : $\ldots$

Eliminating a Redundant Zero-Check

Source Code

$1:\quad \ldots \to 2$

$2:\quad r_1 \coloneqq r_n / r_d \to 3$

$3:\quad r_2 \coloneqq \texttt{test}\ r_d \to 4 \quad \Longrightarrow \quad 3:\quad r_2 \coloneqq \texttt{false} \to 4$

$4:\quad \ldots$

Eliminating a Redundant Zero-Check

	Source Code		Target Code
1 :	$\ldots \to 2$		$\ldots \to 2$
2 :	$r_1 := r_n / r_d \to 3$		$r_1 := r_n / r_d \to 3$
3 :	$r_2 := \texttt{test}\ r_d \to 4$	$\Longrightarrow$	$r_2 := \texttt{false} \to 4$
4 :	$\ldots$		$\ldots$

Eliminating a Redundant Zero-Check

Source Code	Target Code
$1:\ \ldots \to 2$	$1:\ \ldots \to 2$
$\{\mathsf{membij}_{\emptyset}(I) * \mathsf{same\_regs}(I)\}$	
$2:\ r_1 := r_n / r_d \to 3$	$2:\ r_1 := r_n / r_d \to 3$
$3:\ r_2 := \mathtt{test}\ r_d \to 4$	$3:\ r_2 := \mathtt{false} \to 4$
$4:\ \ldots$	$4:\ \ldots$

Eliminating a Redundant Zero-Check

Source Code	Target Code
$1:\ \ldots \to 2$	$1:\ \ldots \to 2$
$\{\mathsf{membij}_{\emptyset}(I) * \mathsf{same\_regs}(I)\}$	
$2:\ r_1 := r_n / r_d \to 3$	$2:\ r_1 := r_n / r_d \to 3$
• $r_d = 0:$ $\{\top\}$	
$3:\ r_2 := \mathtt{test}\ r_d \to 4$	$3:\ r_2 := \mathtt{false} \to 4$
$4:\ \ldots$	$4:\ \ldots$

Eliminating a Redundant Zero-Check

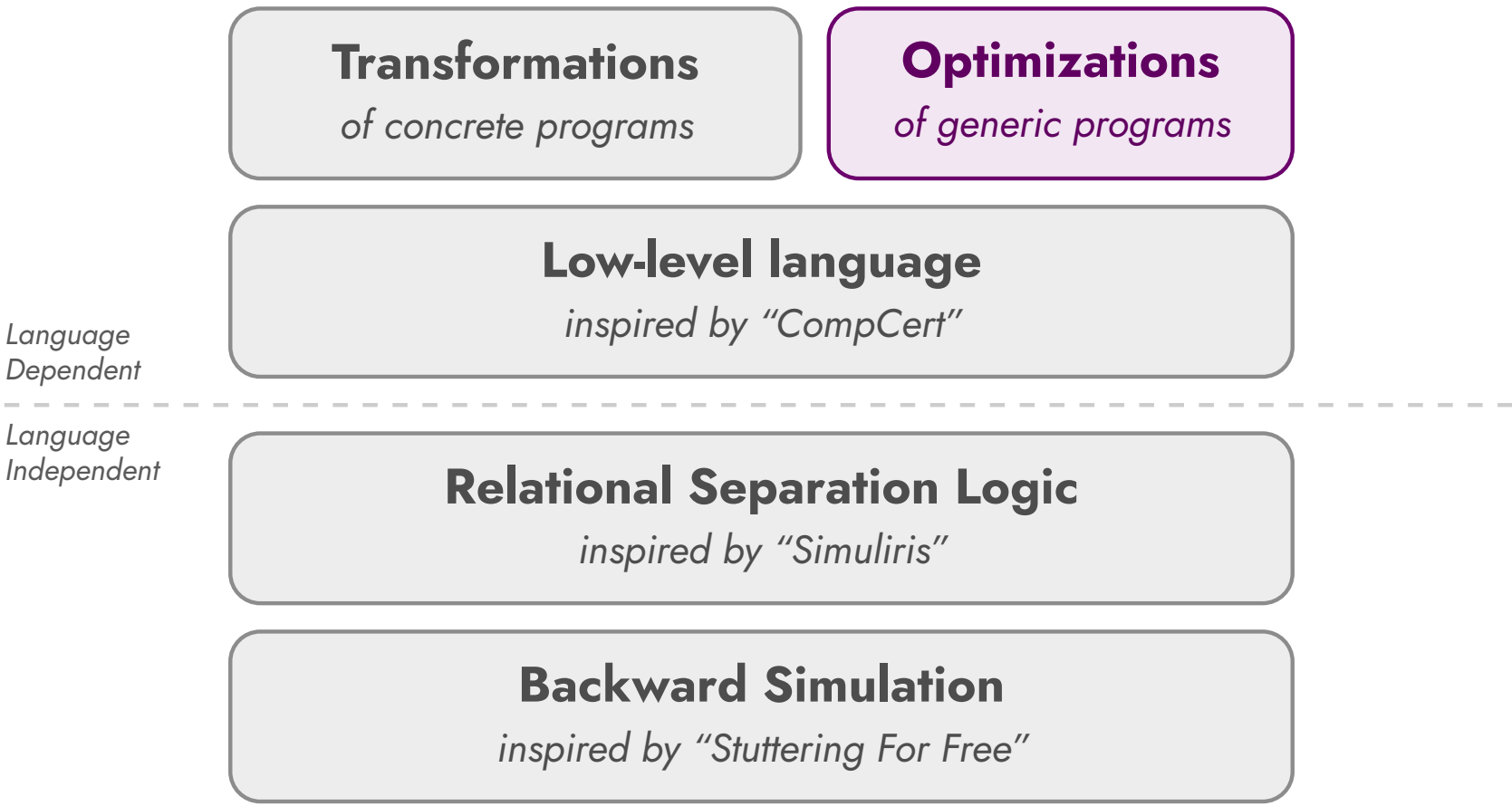
Source Code		Target Code
$1:\ \ldots \to 2$		$1:\ \ldots \to 2$
	$\{\mathsf{membij}_{\emptyset}(I) * \mathsf{same\_regs}(I)\}$	
$2:\ r_1 := r_n / r_d \to 3$		$2:\ r_1 := r_n / r_d \to 3$
• $r_d = 0:$	$\{\top\}$	because $\lightning \in$ source
$3:\ r_2 := \texttt{test}\ r_d \to 4$		$3:\ r_2 := \texttt{false} \to 4$
$4:\ \ldots$		$4:\ \ldots$

Eliminating a Redundant Zero-Check

Source Code	Target Code
$1:\ \ldots \to 2$	$1:\ \ldots \to 2$
$\{\mathsf{membij}_{\emptyset}(I) * \mathsf{same\_regs}(I)\}$	
$2:\ r_1 := r_n / r_d \to 3$	$2:\ r_1 := r_n / r_d \to 3$
• $r_d = 0$: $\{\top\}$	because $\lightning \in$ source
• $r_d \neq 0$: $\{\mathsf{membij}_{\emptyset}(I) * \mathsf{same\_regs}(I) * r_d \neq 0\}$	
$3:\ r_2 := \mathtt{test}\ r_d \to 4$	$3:\ r_2 := \mathtt{false} \to 4$
$4:\ \ldots$	$4:\ \ldots$

Eliminating a Redundant Zero-Check

Source Code		Target Code
$1:\ \ldots \to 2$		$1:\ \ldots \to 2$
	$\{\mathsf{membij}_{\emptyset}(I) * \mathsf{same\_regs}(I)\}$	
$2:\ r_1 := r_n / r_d \to 3$		$2:\ r_1 := r_n / r_d \to 3$
• $r_d = 0$:	$\{\top\}$	because $\lightning \in$ source
• $r_d \neq 0$:	$\{\mathsf{membij}_{\emptyset}(I) * \mathsf{same\_regs}(I) * r_d \neq 0\}$	
$3:\ r_2 := \mathtt{test}\ r_d \to 4$		$3:\ r_2 := \mathtt{false} \to 4$
	$\{\mathsf{membij}_{\emptyset}(I) * \mathsf{same\_regs}(I)\}$	
$4:\ \ldots$		$4:\ \ldots$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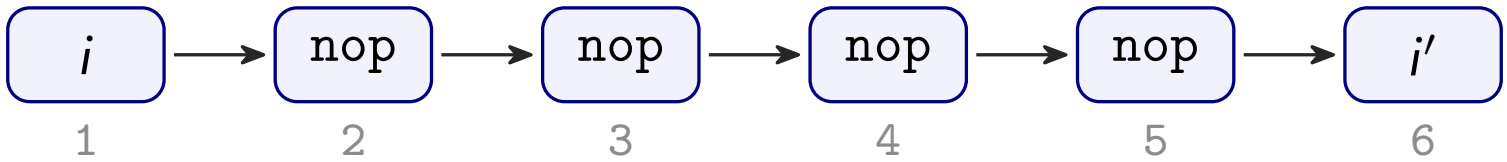
Short-Circuiting Jump Chains



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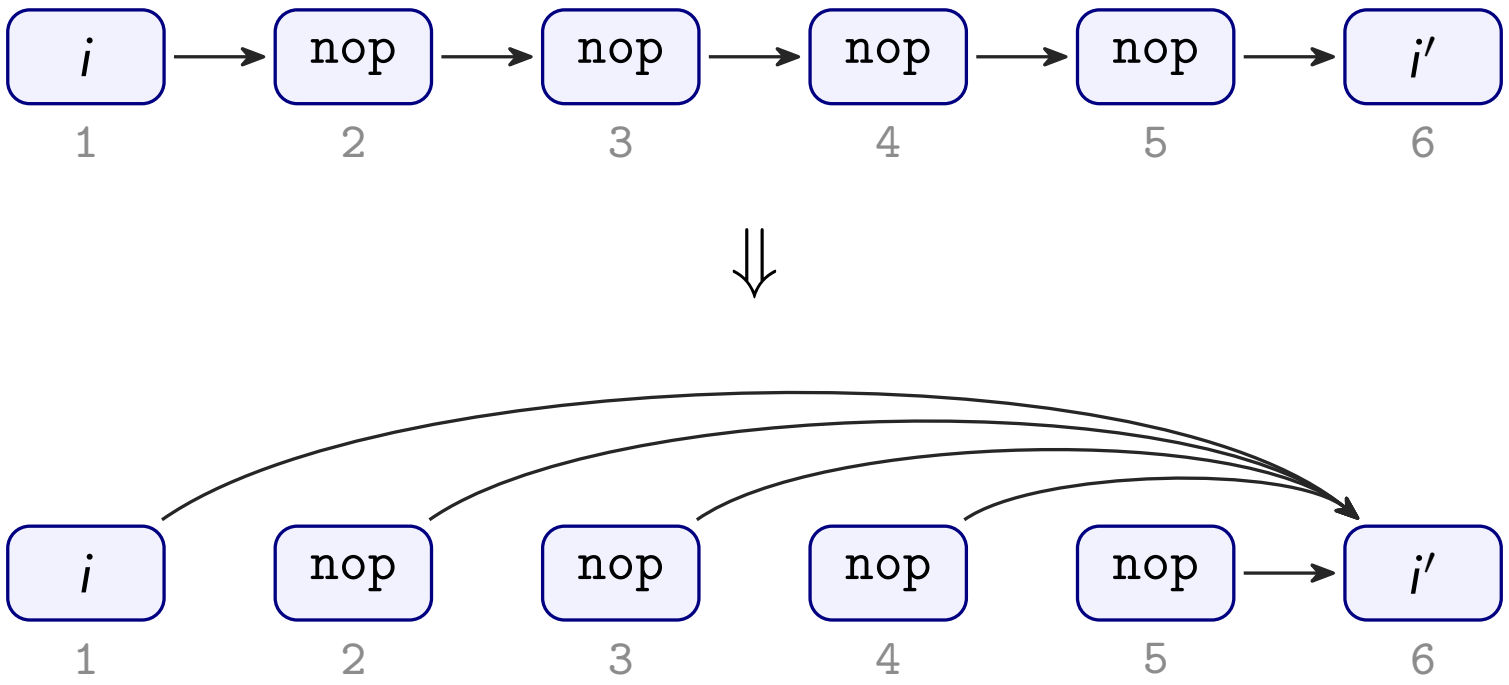
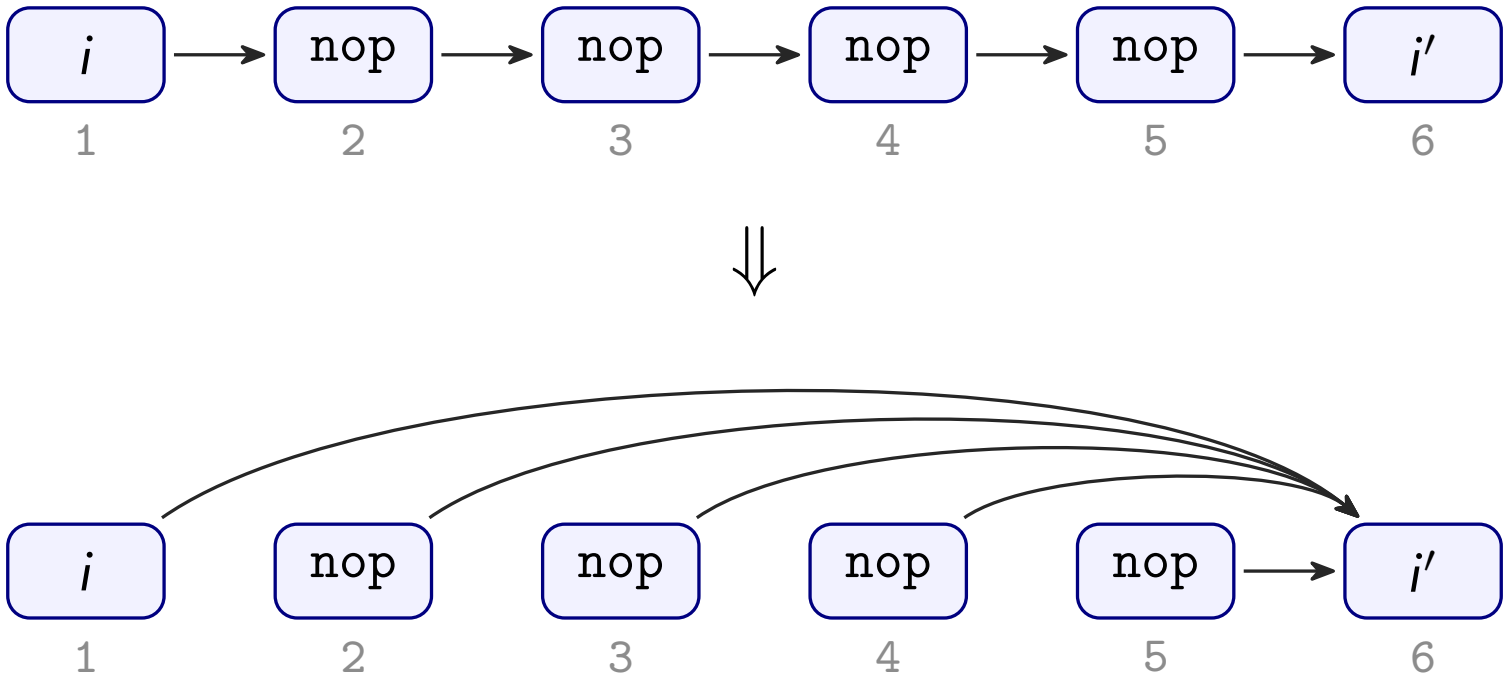
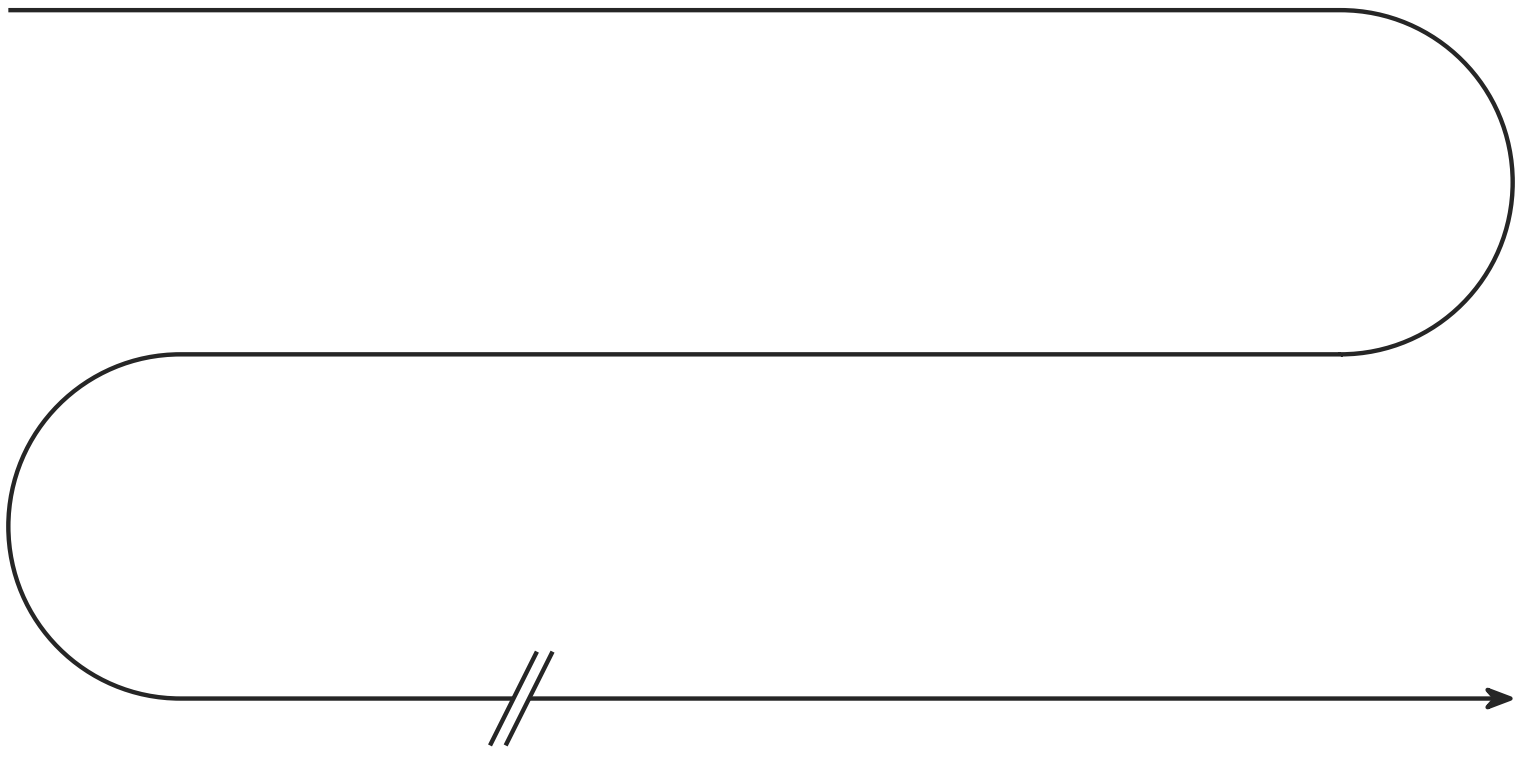


Short-Circuiting Jump Chains

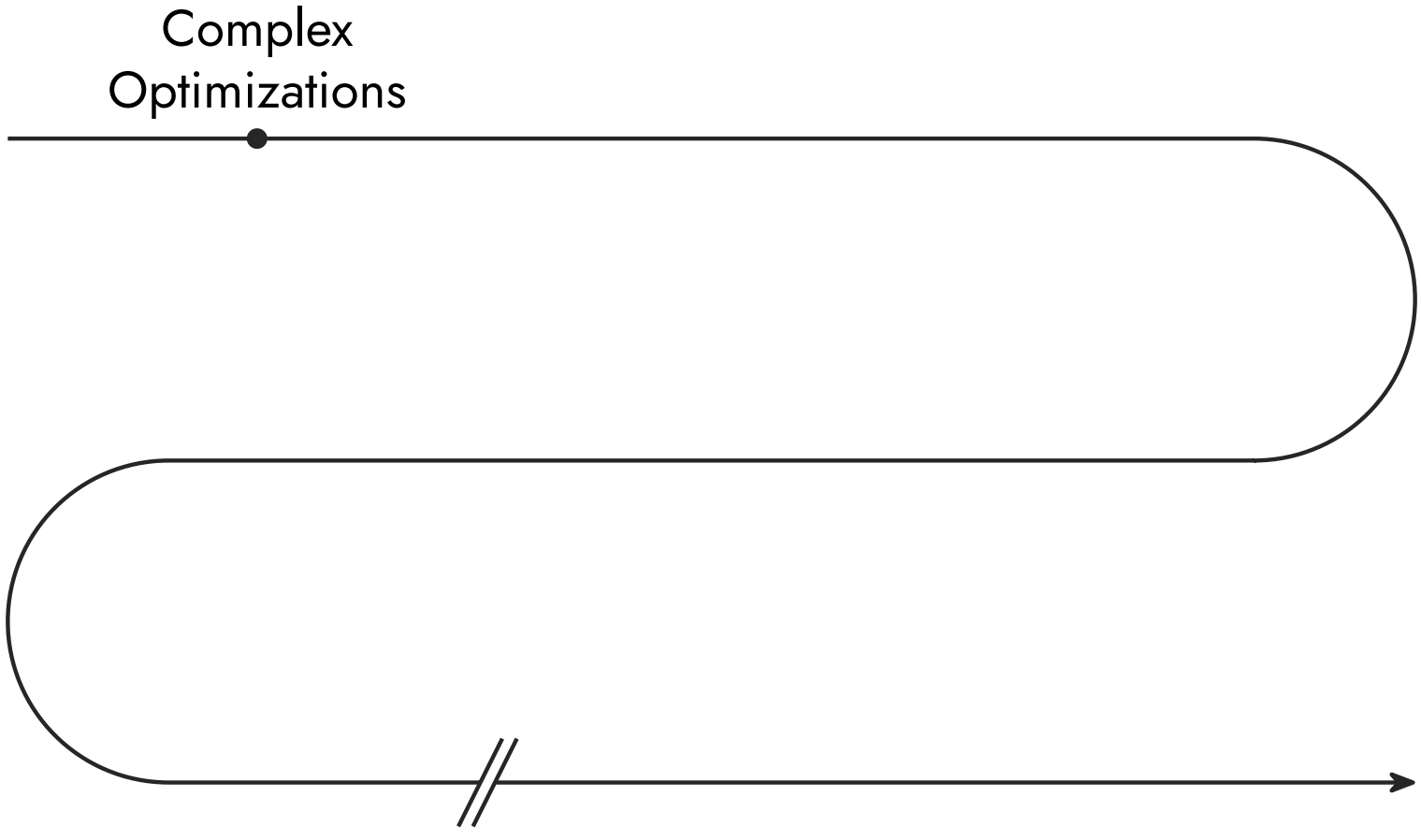


$$\{\text{same\_args}_I\}\ \text{tunnel}_d\,(f) \lesssim f\ \{\text{same\_res}_I\}$$

What Comes Next

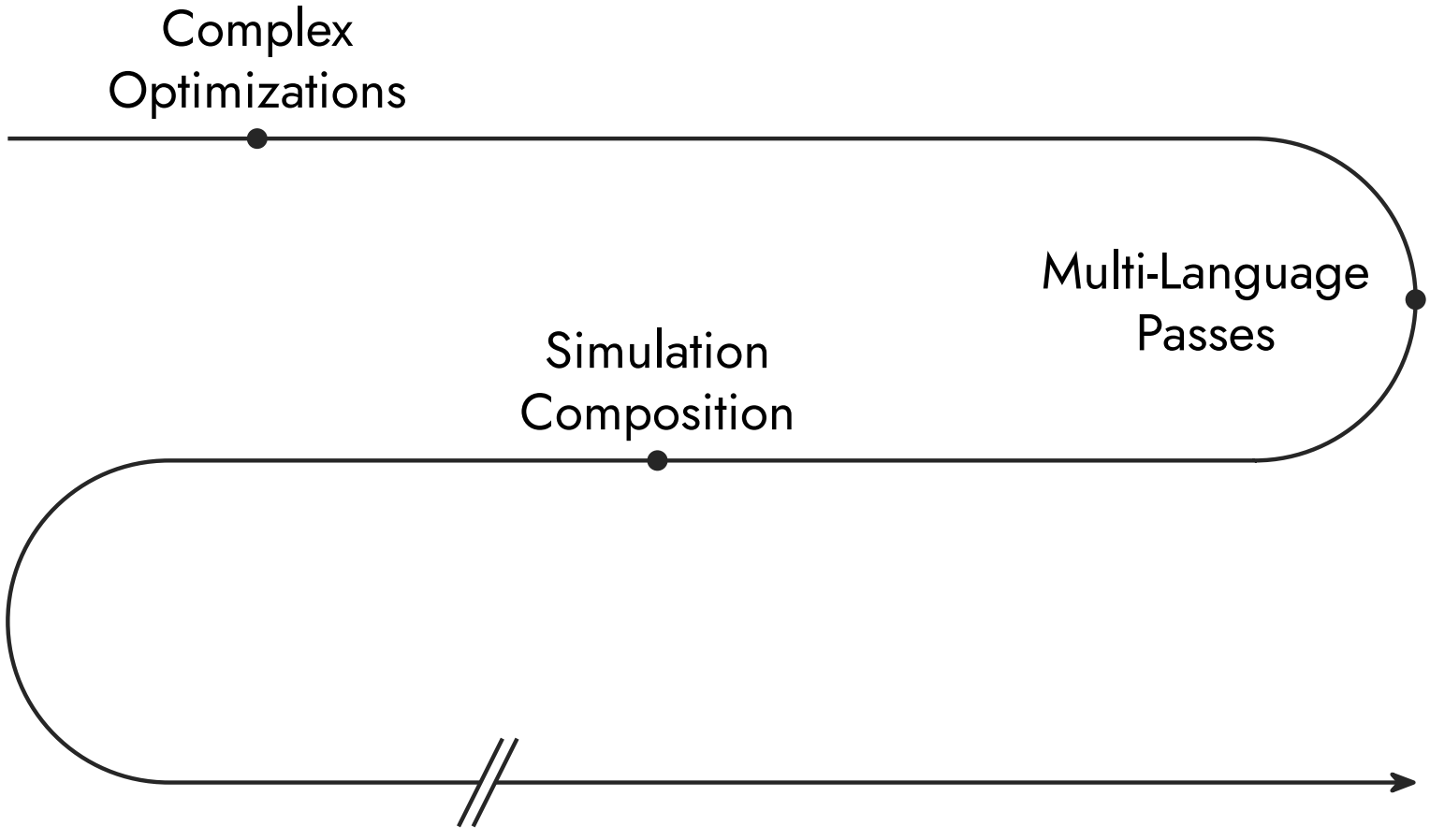


What Comes Next

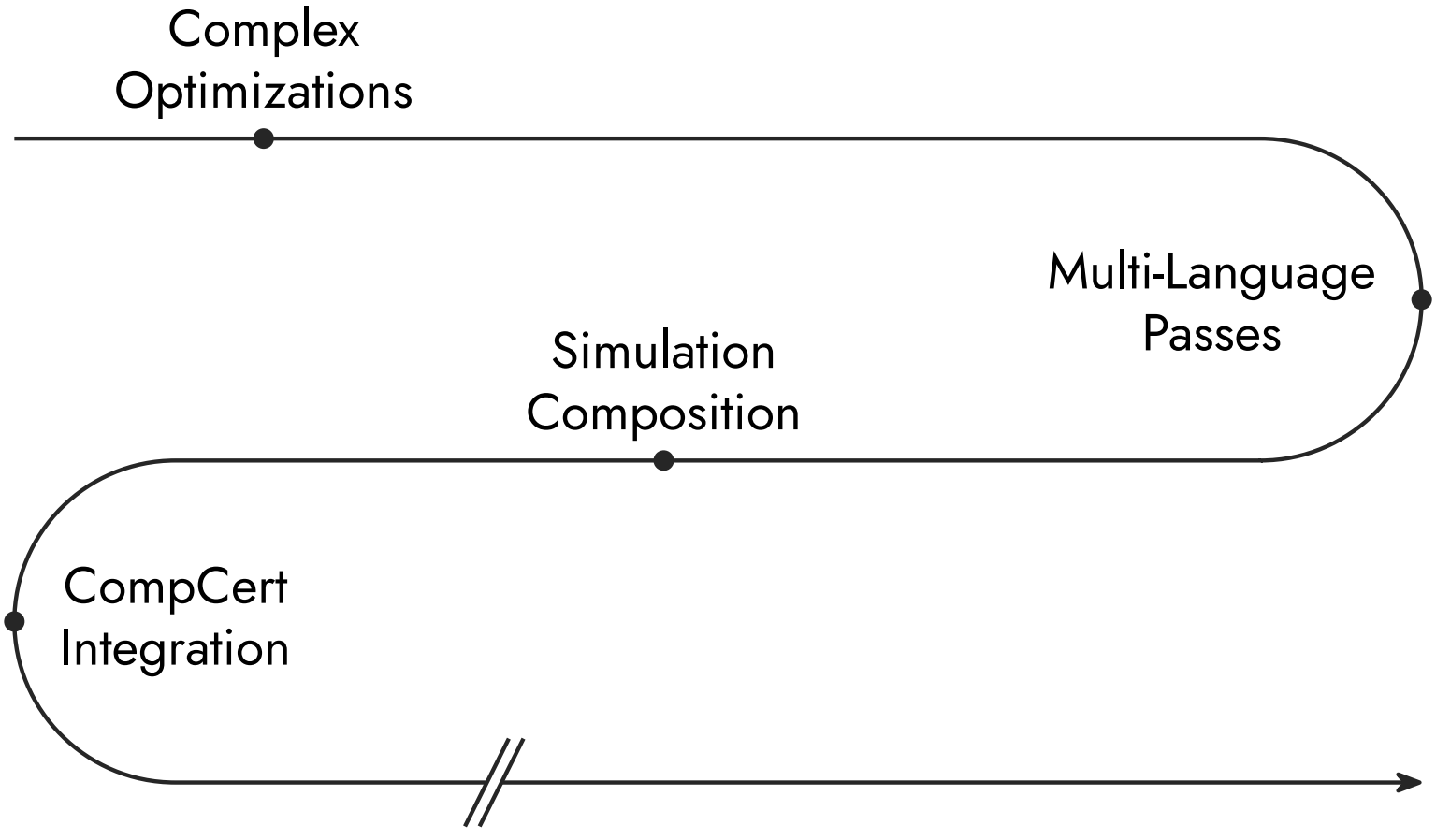


What Comes Next

What Comes Next



What Comes Next



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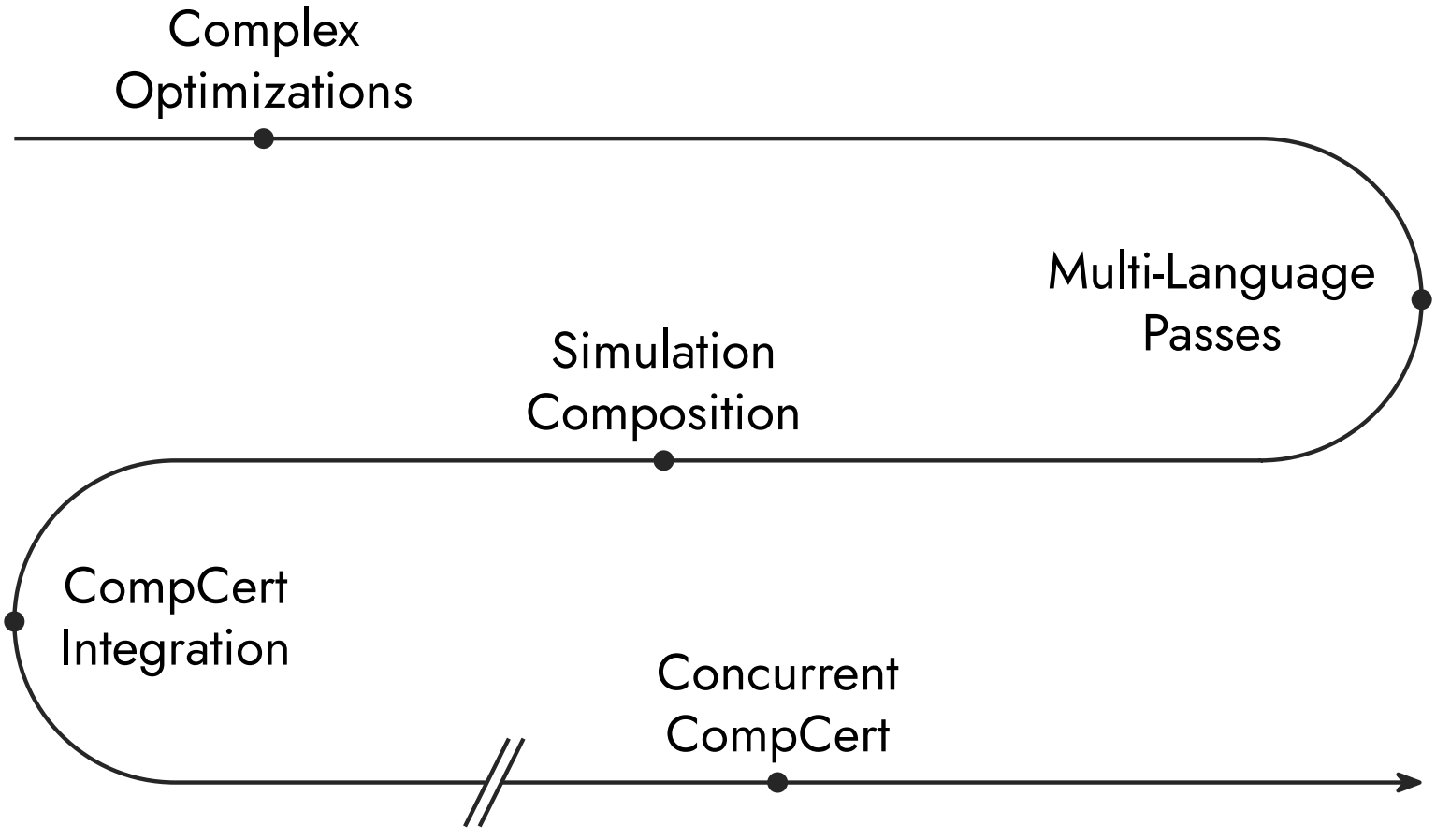

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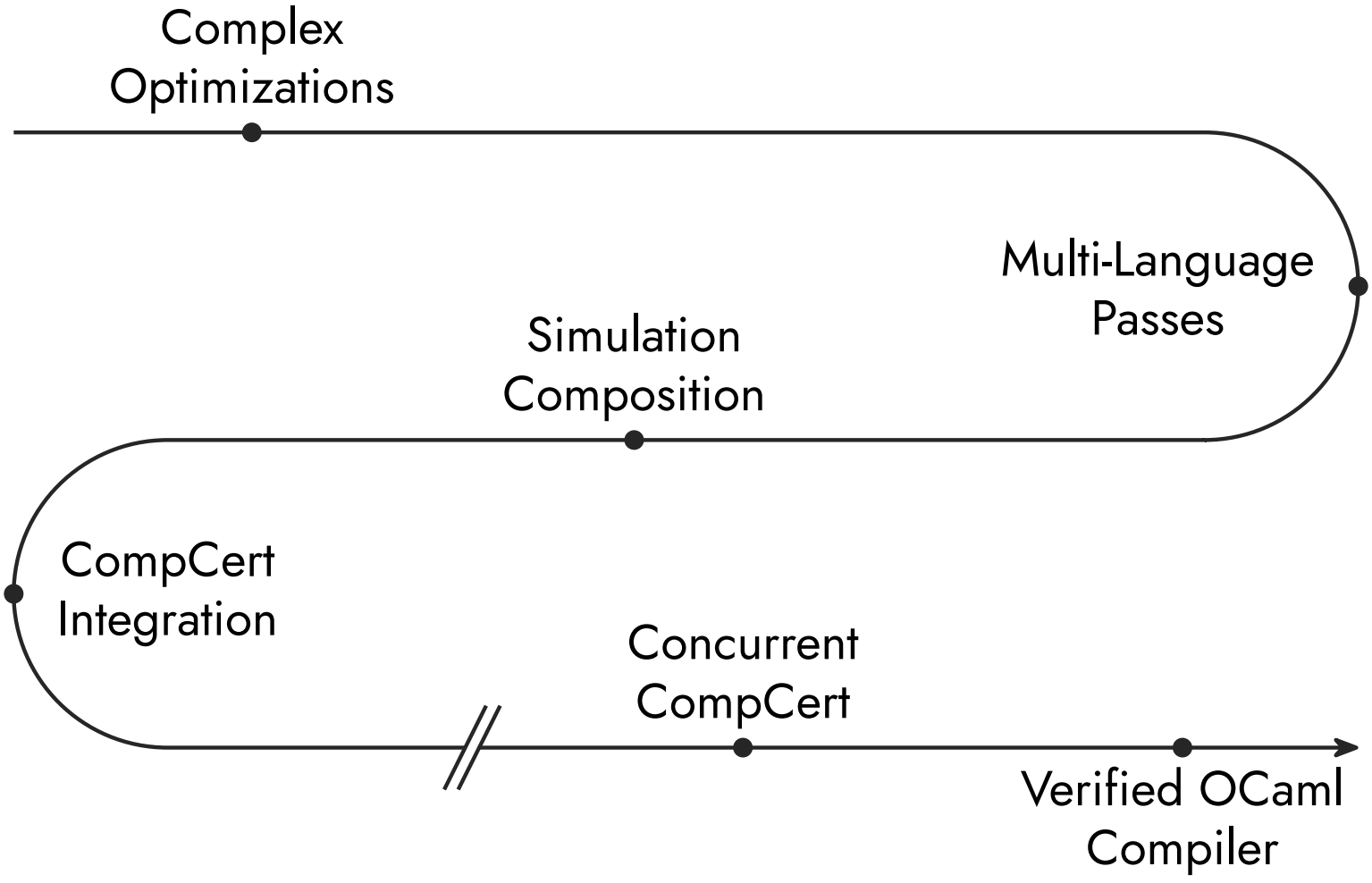
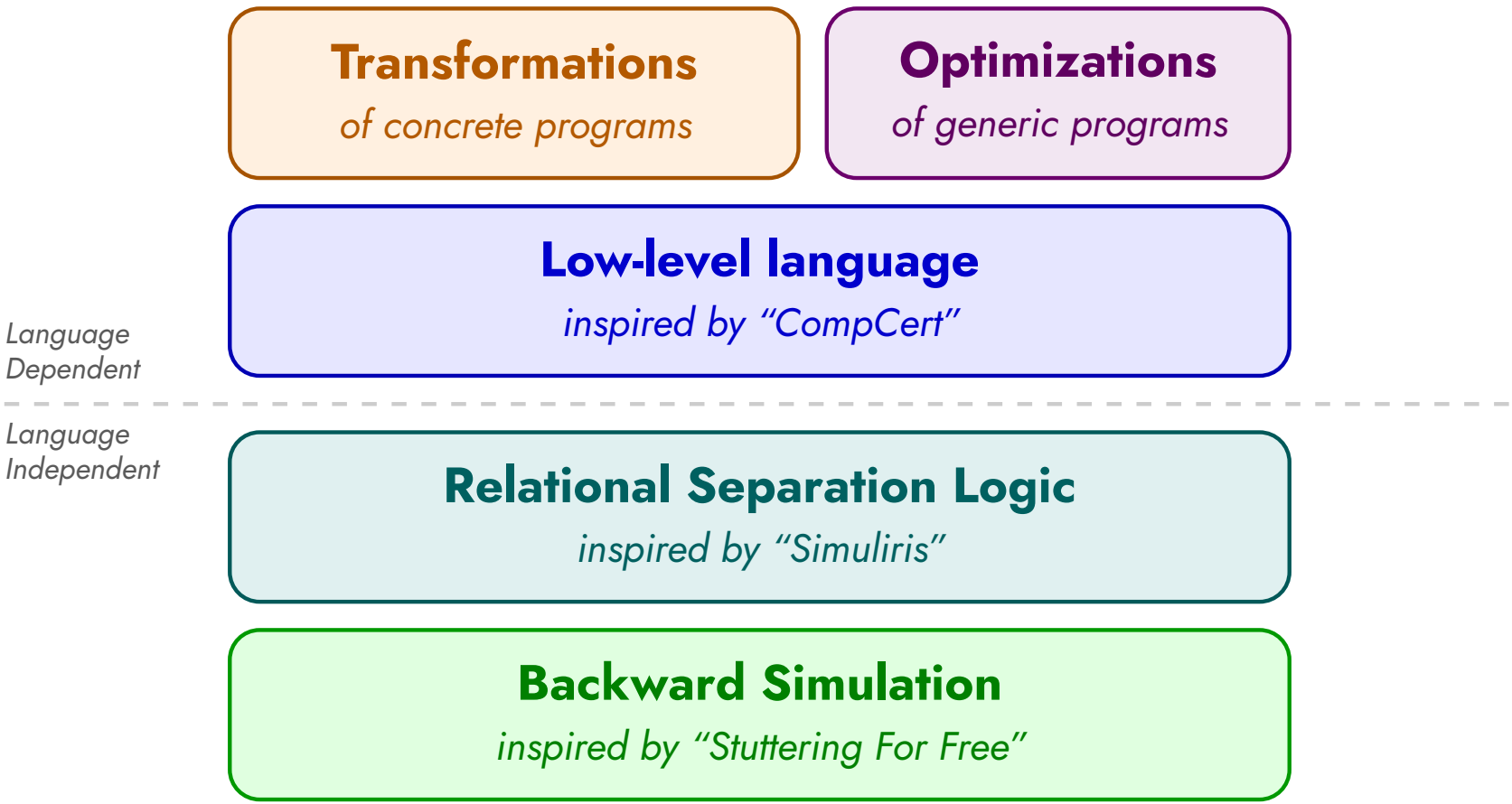


Thanks


Transformations
of concrete programs
Optimizations
of generic programs
Low-level language
inspired by "CompCert"
Language Dependent
Language Independent
Relational Separation Logic
inspired by "Simuliris"
Backward Simulation
inspired by "Stuttering For Free"